



**PHYGITAL GAMIFICATION IN THE EFL CLASSROOM: ADDRESSING  
MOTIVATION AND ENGAGEMENT**

**1048**

THESIS DIRECTOR

UNIVERSIDAD PEDAGÓGICA NACIONAL

FACULTAD DE HUMANIDADES DEPARTAMENTO DE LENGUAS

PROGRAMA: LICENCIATURA EN ESPAÑOL E INGLÉS

BOGOTÁ D.C.

2026

## **Abstract**

This research, aimed to describe the role of Phygital Gamification in addressing motivation and engagement towards English learning among 31 eighth-grade students at Guillermo Cano Isaza school. The study was conducted through a qualitative approach following an action-research design, where classroom observations, field diaries, questionnaires, interviews and artifacts, were used to collect and analyze data. All of this came up after discovering that the participants presented a lack of motivation and engagement towards the English class. The methodology selected for the analysis was the grounded theory, where through a color-coding process, four main categories and 12 subcategories emerged. These categories displayed specific pervasive patterns found during the research. The results indicate that the implementation of phygital gamification increased students' motivation and behavioral engagement, as they displayed a more participative disposition and a positive attitude to the class, and became emotionally involved in the activities proposed. However, the results also showed that this increase in motivation and engagement did not necessarily translate into meaningful or consistent improvements in the linguistic performance of students in the target language. In essence, this research contributes to understanding the effectiveness of implementing gamified strategies in the EFL environment to trigger behavioral aspects such as motivation and engagement and the complex relationship it has with language learning.

**Key words:** Gamification, motivation, engagement, English learning

## Table of content

|                                                        |    |
|--------------------------------------------------------|----|
| Chapter I: Characterization                            | 1  |
| Context                                                | 1  |
| Population                                             | 1  |
| Diagnosis                                              | 2  |
| Statement of the problem                               | 8  |
| Rationale of the study                                 | 8  |
| Research question                                      | 9  |
| General Objective                                      | 10 |
| Specific Objectives                                    | 10 |
| Chapter II: State of the art and theoretical framework | 10 |
| State of the Art                                       | 10 |
| Theoretical Framework                                  | 19 |
| Phyigital Gamification                                 | 19 |
| Motivation and Engagement                              | 21 |
| Grammar Learning                                       | 24 |
| Vocabulary Learning                                    | 25 |
| Chapter III: Methodology                               | 26 |
| Research paradigm                                      | 26 |
| Type of research                                       | 27 |
| Instruments for data collection                        | 28 |
| Observation                                            | 28 |
| Field diaries                                          | 29 |
| Questionnaires                                         | 29 |
| Semi Structured interview                              | 30 |
| Student artifacts                                      | 30 |
| Ethical considerations                                 | 31 |
| Chapter IV: Pedagogical intervention                   | 31 |
| Pedagogical Approach                                   | 31 |
| Vision of curriculum                                   | 32 |
| Vision of EFL classroom                                | 32 |
| Vision of language                                     | 33 |
| Pedagogical intervention                               | 34 |
| Digital Tools                                          | 34 |
| Physical tools                                         | 36 |
| Cycle 1 Building a basis (pre-task stage)              | 38 |
| Cycle 2 Constructing a news program (task stage)       | 39 |
| Cycle 3 Showing my outcomes (post-task stage)          | 40 |
| Chapter V: Data analysis and findings                  | 40 |
| Data analysis procedure                                | 41 |

|                                    |    |
|------------------------------------|----|
| Categories of analysis             | 42 |
| Category 1. Motivation.            | 42 |
| Category 2. Behavioral engagement  | 48 |
| Category 3. Social interaction     | 55 |
| Category 4. Linguistic performance | 60 |
| Chapter VI: Conclusions            | 65 |
| Conclusions                        | 65 |
| Implications                       | 67 |
| Limitations                        | 68 |
| Further research                   | 69 |
| References                         | 69 |
| Annexes                            | 74 |

### **List of tables**

|                                                    |    |
|----------------------------------------------------|----|
| Table 1: State of the art                          | 10 |
| Table 2: Scheme of the cycle 1 of the intervention | 38 |
| Table 3: Scheme of the cycle 2 of the intervention | 39 |
| Table 4: Scheme of the cycle 3 of the intervention | 39 |

### **List of figures**

|                                                                   |    |
|-------------------------------------------------------------------|----|
| Figure 1: Results of the reading exercise                         | 4  |
| Figure 2: Results of the writing exercise                         | 5  |
| Figure 3: Results of the listening exercise                       | 6  |
| Figure 4: Results of the speaking exercise                        | 7  |
| Figure 5: Maslow's hierarchy of needs pyramid                     | 23 |
| Figure 6: Reeve's four interrelated aspects of student engagement | 24 |
| Figure 7: Action research cycle                                   | 28 |
| Figure 8: Categories of the analysis                              | 41 |

## **Chapter I: Characterization**

The following chapter presents in general terms; firstly, a description of the background of the place where the project took place, including institutional and contextual aspects. Secondly, a description of the participants addressed in this study, where features related to their socio-demographics, and English language competencies are included. Thirdly, a brief description of the statement of the problem. Then, the research question, and the objectives proposed to tackle this problem. Fourthly, the rationale of the study.

### **Context**

The CEDID Guillermo Cano Isaza School (GCI) is a mixed public school founded in 1987 which is located in the south of Bogota in the district of Tunjuelito. The school offers education from 1<sup>st</sup> to 11<sup>th</sup> grade in two shifts, morning and afternoon. The GCI school has an emphasis on technical instruction, which aims to the orientation of its educational community to the labor and productive world. Following this, there are three branches that constitute this technical approach, these are: electromechanical design, accounting and finance, and food agroindustry, which are imparted as classes from 9<sup>th</sup> grade in the opposite shift of study.

According to the Institutional Educational Project (Proyecto Educativo Institucional [PEI], 2016), the missionary axis of the school focuses on educating children and youths to be autonomous and responsible in society. This is achieved through the development of academic, technical, and humanistic competencies. What is more, the model of the school follows constructivist theory, which serves as the foundation for all technical specialties (PEI, 2016).

### **Population**

The participants of this research corresponded to 31 teenage students from the grade 802 of the technical public school Guillermo Cano Isaza, of those 16 were girls and 15 were boys. Their ages oscillated from twelve to fifteen years old. These students received their

classes of English within a tight laboratory of languages with internet connection, 10 laptops and a projector with its respective screen. This English class was taken on Tuesdays and Wednesdays from 12:30 to 14:20, which corresponded to the first block of classes. Something to remark about this schedule is that during the class time students were required to abandon the classroom and go to the restaurant to have their lunch provided by the school. This situation altered the time available for the class since they took over 20 minutes of the class to do this activity. Additionally, there was not a given hour for lunch time, so students could be called to go to the restaurant at any moment during the class time.

In order to know about their relationship with the English language, an initial questionnaire (see annex 1) was applied. In the answers collected, it could be found that 57% of the respondents affirmed they liked English while the other 43% disliked it. Nevertheless, 96% considered that learning it was important for their future. Finally, in relation to their attitude towards the English class within the school, 19 of them, which corresponds to the 63% of the sample affirmed that they felt confused during the English class. Moreover 41% expressed that sometimes they had difficulties concentrating during the class and that even when they tried to follow the rules it was usual to get distracted due to boredom. Overall, it could be seen that the learning process of EFL for some students was problematic due to the lack of interest in learning English, the confusion about the topics of the class, and the lack of commitment from them towards the class.

## **Diagnosis**

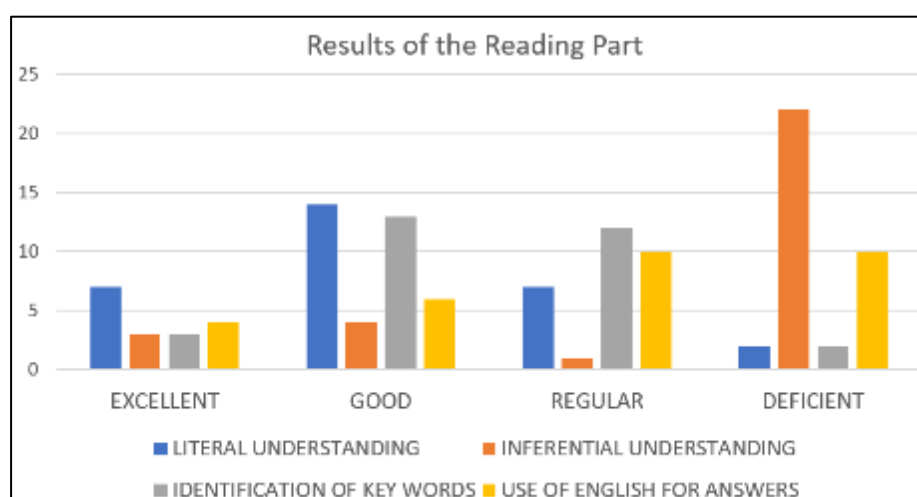
A diagnostic test was developed and applied in the course (see annex 2) with the aim of getting to know the understanding of students' knowledge of English. For the design of this test, the official Foreign Language Proficiency Standards (2017) provided by the National Ministry of Education (in spanish Ministerio de Educación Nacional or MEN) were used, to

create the diagnostic sequence in accordance with the grade level and the expectations for each one of the skills: listening, reading, writing, and speaking.

The first part of the test focused on a reading exercise based on a simple present tense paragraph about the favorite place of a person. The students were asked to answer a series of five questions designed to measure four categories of the reading component: literal understanding, inferential understanding, identification of key words, and writing of answers in English. For the development of this exercise, the specific ability of “extracting general and specific information of a short text written in a simple language” was considered (MEN, 2017, p. 22). The results showed that 46% of the students achieved the literal level with a score of good in the scale, which meant that they could extract some of the main ideas of the reading. In terms of inferential understanding, 73% of them got an insufficient score as they did not offer any inference about the reading but just answered the question using yes or no.

Regarding the identification of key words, 43% of them were able to write several key words of the reading, while the other 40% wrote at least one key word. Finally, in the task of writing their answers using the simple present tense in English, 6 % of the sample achieved a score either regular or deficient, which meant that they had difficulties when using the grammar tense properly to create short sentences. Overall, the results showed that the participants of the study had a decent level of reading comprehension in terms of literal understanding and identification of key words. The low results in the inferential level could be related to the fact that an instruction of writing an assertion instead of just answering the question was missing.

Figure1: Results of the reading exercise

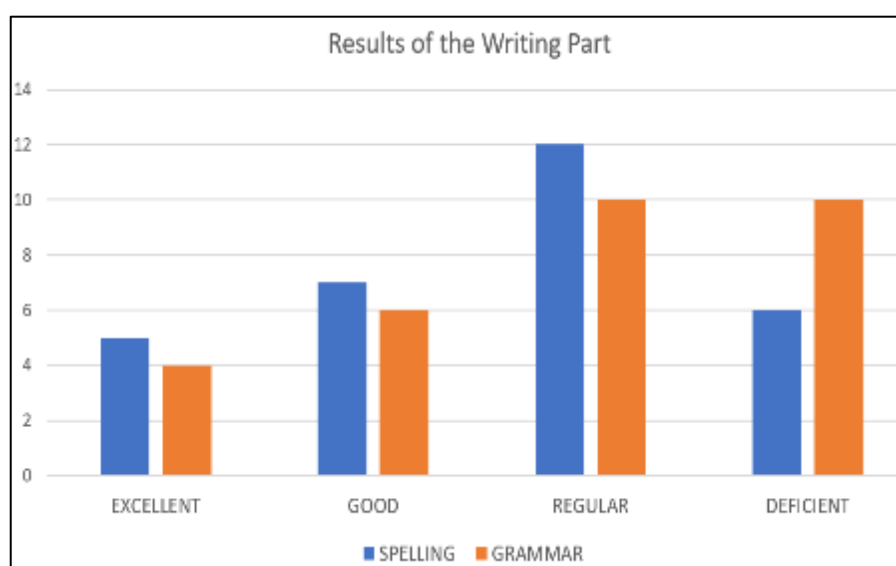


The second section (writing production) was based on the specific ability mentioned in the national policies of education of “writing a short text based on the environment of the student and the familiar objects present on it” (MEN, 2017, p. 23). It is crucial to mention that in the planning of this section, the creation of a short paragraph, in which students mentioned their favorite place, was considered. Aspects such as spelling, use of vocabulary, grammar and coherence in their writing task were taken into account for the diagnosis. Nonetheless, this task could not be undertaken due to the constraints with the time of the classes. Therefore, for the analysis of the writing production the responses of the students in the reading section were used, which were examined through two categories: Spelling and grammar.

The results exemplified that while 33% of the responses presented either an excellent or good level in terms of grammar when writing, the other 66% presented a regular or deficient level, which meant that the students were susceptible to making written grammar mistakes. In terms of spelling, it did not vary that much, as the results provided showed that most of the students (60%) were prone to make spelling mistakes, while the other 40% showed either a good or excellent level in spelling when writing words.



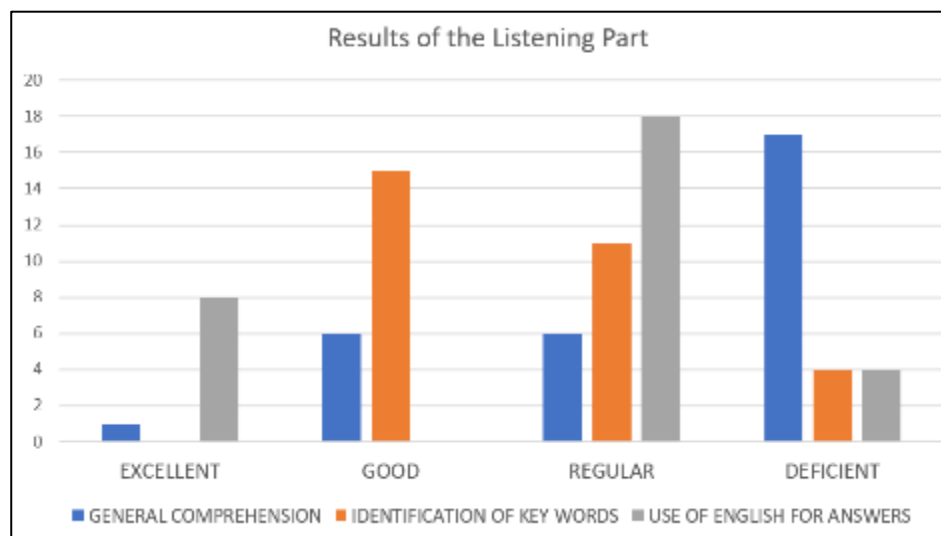
Figure 2: Results of the writing exercise



The third part of the diagnosis was intended to assess the listening comprehension of the students in terms of three categories: general listening comprehension, recognition of key words and writing of answers using English. It was based on the standard that indicates that a student from this grade should be able to “comprehend an oral description about a place or a person” (MEN, 2017, p. 22). In that line, an audio of a description of the favorite place of a person using the simple present tense was created.

The results showed that in terms of general listening comprehension most of the students, equivalent to a 40%, presented an insufficient score, which meant that their comprehension was too limited when it came to listening and therefore, they were only able to regain certain pieces of information from what they listened to. In relation to their identification of key words, 50% of them were able to identify certain words that appeared in the exercise, although those words were isolated in many cases without sentences that connected them, which could be proved with their overall low results in the use of English with the 73% of the responses in the regular and deficient score.

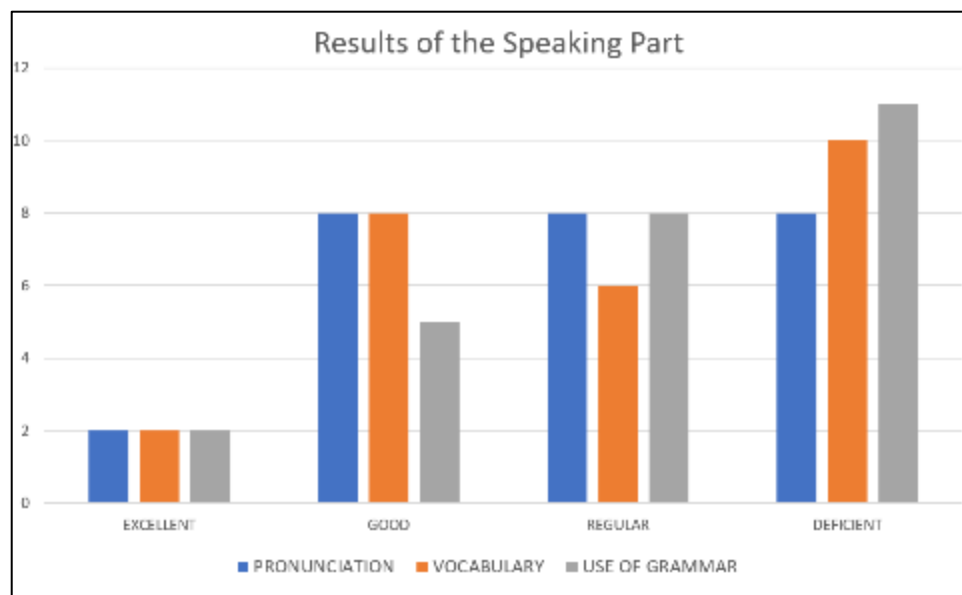
Figure 3: Results of the listening exercise



The fourth and last part of the test focused on oral production, and it consisted of a short exercise in which the students had to give a brief oral description of their best hobby or favorite activity in life using three words related to this topic given by the pre-service teacher. This was based on the specific ability of making descriptions with simple sentences of a person or a place mentioned in the standards provided by the national ministry of education (MEN, 2017).

The findings showed that 61% of the respondents presented difficulties when pronouncing several words in English, and even some of them used a Spanish pronunciation for the words given. In terms of vocabulary, again 61% of the sample presented problems using different vocabulary rather than the words given to create the oral description, which displayed the lack of vocabulary that a big proportion of the population had, being this one of the biggest language insufficiencies. Finally, the use of grammar orally evidenced that 73% of the participants had a low level of use of grammar structures to convey their ideas, which was noteworthy since grammar was the main aspect worked by the head teacher during the classes observed. The fact that there were deficiencies in terms of grammar when there was input of it in the class could also imply the lack of interest that the students had in the class and in the teaching methods used by the headteacher.

Figure 4: Results of the speaking exercise



During the four stages, a lack of vocabulary was demonstrated in the students while performing the different exercises proposed. As was outlined in the reading section, although at least 43% of the students were able to use key words for completing the writing exercise, these key words were already written in a paragraph, and the students just copied the sentences exactly in the way they appeared without using other words, linking connectors, or even prepositions. This showed the lack of proficiency in English vocabulary that the population had.

Furthermore, in the listening exercise, students were asked to write down on paper sentences about what they understood from an audio. Even when 50% identified at least one word, these words were written in an isolated way without linking words that could convey a complete idea. Again, this evidenced the lack of vocabulary that the students had when creating whole sentences. In addition, a similar result occurred in the speaking section, where the students had to produce an oral sentence based on three words that the researcher offered them in relation to their interests. 73% had problems creating these sentences, since they did not find other words to connect the given words or even use different words from the given ones.

Moreover, in this exercise problems regarding word pronunciation were evidenced, since Spanish pronunciation was still prevalent.

### **Statement of the problem**

By observing the dynamics displayed by the students during five sessions and the data recorded in different field notes, it was found that in general terms more than 54% of the participants presented a lack of motivation and engagement towards the English class in their school, which could be associated as the catalyst of subsequent problems that were found as well, overall related to language performance. For instance, the lack of commitment to the class was really evident since several students preferred to dedicate the time of the class to do other trivial things like chatting, playing games or using their cell phones instead of paying their whole attention and disposition to the class. Besides, the overall low level of English proficiency shown on the classes observed and the diagnosis test results in each one of the skills, particularly use of grammar and vocabulary.

These aspects could be directly related to their relationship with the language during the class, as the questionnaire also revealed that 63% of the students felt confused during the lessons. This aspect suggested that the student did not understand all the input provided by the teacher or the purpose in learning it. This lack of understanding may have contributed to a decrease in their motivation and at the same time, it increased their distraction during class time. This problem could also be related to the way in which the lessons were taught, as the teaching approach used by the head teacher tended to be traditional and mostly unidirectional, which limited students' active participation and reduced their opportunities to engage in the language.

## **Rationale of the study**

To address the lack of motivation identified in the classroom, the present study contemplated the implementation of a pedagogical intervention mediated by phygital gamification with the aim of making the learning process of the target language through grammar and vocabulary more appealing, more participatory and contextualized for students, triggering as a consequence, their motivation and engagement towards the class.

As it became necessary to approach the teaching practices of English in the classroom following a different method with the aim that students felt more motivated and engaged to learn the target language, phygital gamification emerged as the medium to allow these features to arise. As Valenzuela (2021) pointed out, gamification has the potential “to solve the constant problems of motivation that affect education” (p. 93). With this in mind, the initial focus of the study was to shift the classical teacher-centered approach to turn the class into an interactive learning environment where students could co-construct knowledge through their participation in the activities proposed. Therefore, from an institutional perspective, the aim was to renovate the teaching practices in the Guillermo Cano Isaza school within English as a foreign language. In this sense, this research could become a model to inspire teachers at the school to find different teaching strategies to conduct their classes, by putting a spotlight on the students’ active participation rather than in the role of the teacher as the facilitator of information.

Likewise, from a national perspective, this research aimed to show possible repercussions of implementing phygital gamification as a pedagogical tool when teaching English as a foreign language, offering a different method from the traditional workbook-based and teacher centered practices, which are still imperative in the educational settings due to the concerns imposed by the educative policies that aim to the accomplishment of certain levels of proficiency in English lied on the Common European Framework of Reference for languages (CEFR), to insert the country into the globalized dynamics.

### **Research question**

- What is the role of Phygital Gamification in addressing motivation and engagement towards learning English within the classroom in the 702 students at the GCI school?

### **General Objective**

- To describe the role of Phygital Gamification in addressing motivation and engagement towards English learning among 702 students at GCI School

### **Specific Objectives**

- To identify how Phygital Gamification triggers students' interest in English learning in the class.
- To examine how Phygital Gamification influences students' behavioral engagement in the classroom, particularly in terms of participation, attention, and task involvement.
- To determine the extent to which Phygital Gamification contributes to students' linguistic performance in English throughout the intervention period.

## **Chapter II: State of the art and theoretical framework**

The following chapter introduces a series of studies that were considered in advance for the creation of the current project. Moreover, the main concepts that underpin and provide the theoretical axis of the research are presented.

### **State-of-the-Art**

Nine studies that held a closeness to the present research were selected. At the moment of finding related documents in several databases, the aim was selecting studies that first had a qualitative design, since the present study pretended to use this approach. Second, had a similar population, and third, held a focus on the use of gamification for teaching English. At the end, nine studies from five institutions were considered for this research from local, national and

international contexts. These studies shared the issue of gamification implemented into educational settings. Also, they emphasized why the use of it could be beneficial when approaching the process of teaching English as a foreign language.

*Table 1: State-of-the-art*

| <b>Title of the study</b>                                                                                                                                                | <b>Author</b>                                      | <b>Affiliation</b>                                        | <b>Year</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------------------------|-------------|
| Gamification: A Tool to Improve Speaking Skill Through the TBL Approach.                                                                                                 | Ana Almanza                                        | Universidad Pedagógica Nacional (Colombia)                | 2024        |
| Gamification, a strategy to enhance speaking skills in a blended learning environment.                                                                                   | Luisa Guerrero                                     | Universidad Pedagógica Nacional (Colombia)                | 2023        |
| Gamification: a mission to foster students' engagement and interaction in the efl classroom                                                                              | Laura Esquivel                                     | Universidad Pedagógica Nacional (Colombia)                | 2019        |
| Gamification as a tool for practicing speaking skills                                                                                                                    | Laura Ramos<br>Laura Herrada                       | Universidad Distrital Francisco José de Caldas (Colombia) | 2023        |
| Estudio contrastivo del efecto motivacional de tres estrategias formativas virtuales en el contexto de la preparación del examen nacional Saber 11° en el área de inglés | Estefania Matallana                                | Pontificia Universidad Javeriana (Colombia)               | 2017        |
| La Gamificación como elemento motivador en la enseñanza de una segunda lengua en educación primaria.                                                                     | David Gonzalez                                     | Universidad de Burgos (Spain)                             | 2017        |
| Gamificación en la enseñanza del inglés como lengua extranjera en educación superior                                                                                     | Johnatan Martínez<br>Mónica Riaño<br>Gloria Suárez | Universidad El Bosque (Colombia)                          | 2024        |
| The impact of gamified learning using Quizziz on ESL learner's grammar achievement                                                                                       | Anh Tuan Pham                                      | FPT University (Vietnam)                                  | 2023        |
| The impact of gamification on secondary school students' grammar proficiency                                                                                             | Gözde Koç Selim<br>Soner                           | Başkent University (Turkey)                               | (2023)      |

In the first place, the project *Gamification: A Tool to Improve Speaking Skill Through the TBL Approach* carried out by Almanza (2024) examined how the use of gamification as a

pedagogical strategy could improve English-speaking skills in seventh-grade students, since the researcher found that oral production was not promoted by the teacher in the classroom and therefore the students had a low level of it. For the sample of this study, it consisted of 34 teenage students from the CEDID Guillermo Cano Isaza School. As for the design used for this study, it was an action research approach. The intervention proposal drawn upon a series of tasks based on the creation of a series of descriptions of horror characters that had to be done in a written and oral way, while the students received points after each outcome they presented.

The results concluded that the strategy increased participation among the students and their confidence toward the oral production in English as well. It also improved their pronunciation skills as they lost fear of making mistakes. This research was significant to the present study for two main reasons. First, it was carried out within the same setting and with a similar population of the present project, which was the CEDID Guillermo Cano Isaza in an EFL class with seventh graders as well. Second, as it was shown in the results, this study exemplifies how adapting teaching practices to students' interests can boost their learning by promoting their motivation towards the English class. This study illustrated how the use of digital presentations such as Canva, and contextualized use of English in creating descriptions of students' home is beneficial for addressing language in the classroom in a familiar way for students.

The main differences of this research in contrast with present study lie in the language approach and the scope of gamification. While the first focused on addressing individual oral production, so that student presentation activities were planned in this line, the present study focuses on grammar and vocabulary use, in short collaborative exercises. Moreover, the gamification implemented in the first study was reduced to using points and horror themes as part of the class instead of creating a gaming environment where elements such as avatars, awards, challenges, as the present study did.



In the second place Guerrero (2023) conducted the action-research project *Gamification, a strategy to enhance speaking skills in a blended learning environment* aimed to promote speaking skills in English by using gamification as a pedagogical strategy in a population made of ninth graders of the IPN school (Instituto Pedagógico Nacional). The pedagogical intervention consisted of the creation of an imaginary “space shift” by the teacher where the students learned grammar, vocabulary and expressions while they completed missions and challenges. Moreover, the use of the app ELSA was considered as part of the autonomous work of the students.

The conclusions pointed out that the speaking skills were improved during the intervention, as well as the interaction with the target language through gamification. The contributions of this study to the present one lie in how creating a gaming environment influences students’ class performances and commitment towards class. Regarding the differences between Guerrero’s research and this one, the first one was carried out in a non-presential way due to the Covid-19 pandemic. Therefore, the ICT tool was applied in semi-presential environments, rather than in a face-to-face one as the present study contemplated to. Moreover, the focus on Guerrero’s study was oral production rather than vocabulary and grammar learning.

In the third place, the research *Gamification: a mission to foster students’ engagement and interaction in the EFL classroom* conducted by Esquivel (2019) pretended to determine the impact of using gamification as a pedagogical strategy to make the EFL classroom an engaging environment for 30 fourth graders in the Domingo Faustino Sarmiento School. The proposal was centered on a series of lessons in which the students had to consider the class as a gaming environment where they completed challenges and missions while they learned new vocabulary by collaborative work with others. The proposal dwelt upon student interaction as the researcher identified the problem of lack of interaction among themselves when it came to

use of English in a practical way that in her words was due to “the lack of an involving, engaging, and motivating environment” (p, 3).

The findings showed that this strategy was overall useful, since after the intervention the students were able to interact with unknown people in English participating in short conversations. This research was significant to the present study since it established a positive relationship between gamification and student engagement within the educational setting, considering that the students expressed that they felt enthusiastic, interested and lively through the development of the intervention sessions. Although there are some similarities of this research conducted by Esquivel with the present study, the main difference lies in the use of digital tools to create a gamified environment. While Esquivel used physical elements such as cards of coins, badges, and score cards to implement the gamification strategy, the present study conducted considered the use of digital and physical elements simultaneously to apply gamification.

In the fourth place, the research *Gamification as a tool for practicing speaking skills* conducted by Ramos and Herrada (2023) at the Universidad Distrital, intended to determine how games used as a learning strategy could impact the speaking skills of 61 tenth graders in a public school in northeast Bogotá. The pedagogical proposal by the researchers was formed of a series of lesson plans that linked grammar input with activities through the task-based approach. Such activities were focused on enhancing the collaborative work and the oral production, since with a diagnosis they found that “the students had difficulties with oral communication skills, particularly with fluency” (p. 12).

Moreover, they identified socio-affective problems related to self-confidence in the students when they were about to speak. Some of the activities implemented were crime cases, charade surveys carried out by the students and implemented among them, and descriptions of their favorite celebrity. This improved students’ skills in terms of speaking, such as grammar,

pronunciation, vocabulary and fluency. Likewise, other aspects were improved such as the motivation that students had in their learning process and the collaborative work among their fellow students.

The contribution of Ramos and Herrada's research lies in the importance of proposing activities that allowed students to address their context, such as creating description of the places they lived in. The main difference lies in the gamification conducted in both studies, while Ramos and Herrada's study lied in the use of games and activities to conduct the class, this study strived to create a gaming environment where the mechanics of games are adapted to the class beyond using games already created into a class.

The fifth study titled *Contrastive study of the motivational effect of three virtual teaching strategies in the context of the 11<sup>th</sup> grade national exams in English* by Matallana (2017) was considered for the present project. The purpose of this research dwelt upon contrasting three virtual teaching strategies and the effect they had on motivation for students towards learning English in 11<sup>th</sup> grade. This objective was stated after the researcher found through a documental revision that the national exams in English were not related with the contents of practices that the teachers tended to work with at school in 11<sup>th</sup> grade, this leading to frustration for students who did not reach the desired results and were forced to take other study alternatives such as mock up national exam courses where the teaching practices are intensive in a short period of time, and shapes the perception of students from the language, making them feel unmotivated.

The population lied on 26 11<sup>th</sup> graders from a school in Bogota who were being trained to present the national exam. The intervention consisted of the use of three formative strategies conducted through the ICT tool Tutorya. The first one was a series of questionnaires developed by the teacher in the platform that students completed weekly. The second was a series of formative videos hosted in Tutorya that students could watch at any time to reinforce their

understanding of the topics, and the third was classes online conducted through the same platform which could be recorded to be watched later.

In relation to the results, the author concluded that the strategies implemented, especially the questionnaires and the online classes, contributed positively to the motivational and leaning processes of the students, since they showed a progress over the topics they should undertake in their way to the exam, and displayed a committed willingness to use the platform and the tools. Matallana's study contributed to the present research in displaying how the use of ICT devices and tools can be beneficial for creating attractive didactic material for students, which contributes to enhancing their learning process. In relation to the differences between the two studies, they lie in the fact that Matallana's study conducted the intervention in a virtual scenario while this study was conducted in a face-to-face scenario using digital and physical tools.

The sixth research considered for this literature review was conducted by Gonzalez David from the University of Burgos in Spain in 2017. The study was titled *Gamification as the motivator element in second language teaching in primary*, translated and aimed to comprehend the impact in motivation while using digital applications and websites that could enable gamification in the English class. This objective was proposed after discovering that the students presented a lack of concentration and attention to the class. The population for this study lied on 24 primary students from second grade in a school in Burgos, Spain. His pedagogical proposal consisted of the use of two digital applications, Plickers and ClassDojo. The first one was used with the purpose of creating traditional questionnaires while the second was used for creating a system of points. Along with the implementation of these two tools the students had to complete some exercises from a traditional book.

Regarding the results, the intervention achieved the purpose of increasing motivation towards the class, the students showed interest in completing the activities since it provided

them with a different perspective of the class rather than traditional practices, thus, fostering their efficiency consequently. Moreover, the intended contents of the class were completed in a proper way. This study contributed to the present one, with the digital tool of ClassDojo, which allows the teacher to create a virtual classroom where students can obtain class points. The main differences lie in the absence of correlation elements of these points in a physical way, as well as the lack of presence of other elements such as badges, timers, avatars and challenges that help create and consolidate the gaming environment.

The seventh research considered for the state of the art was *Gamification in teaching English as a foreign language in higher education* conducted by the researchers Martinez, J., Riaño, M., and Suarez, G., (2024) as part of their degree work at El Bosque University in Bogotá. The purpose of this research focused on formulating a series of orientations for implementing gamification in EFL teaching at National University. The researchers analyzed the importance of exploring different teaching techniques that could foster the didactic sphere in environments where it is scarce and nonexistent, and where the predisposition to learn English tends to be negative, as it was in their target population.

This study worked with students from the program of foreign language (PLE) in the levels I and II of the National University of Colombia. This study was documentary analysis research where the researchers focused on analyzing literature about gamification and teaching English as a foreign language to describe the most relevant characteristics when approaching gamification. At the end, the researchers established 5 main categories and 6 subcategories to be considered when conducting this teaching technique: Motivation, pedagogical objectives, constraints, characteristics of gamified activities, sources and tools.

The results drew certain benefits and drawbacks of using gamification in the EFL teaching process. This study contributed to the present one by exhibiting how motivation can be stimulated in learners with the use of gamification, making them be more participative and

involved in the teaching activities. Also displaying how through collaborative work, communicative skills can be improved significantly. No less important, showing how tasks such as grammar, understanding vocabulary and memorization can be achieved after engaging students to satisfy their needs of competence and awards. On the other hand, this research also provided suggestions on how gamification should be implemented properly into a class to avoid negative consequences in the learning process, anxiety and frustration in learners with a low proficiency level.

The eighth research was conducted by Pham (2023) in Vietnam, and it was denominated *the impact of gamified learning using Quizizz on ESL learners' grammar achievement*. The objective in this research was to examine a possible impact by implementing gamified learning using the website Quizizz in an English as a second language class. The participants of the study were 63 university students whose ages went from 18 to 19 years old and who were taking grammar English classes at university. The whole sample was divided into two groups, the first group had 33 students, and it was the experimental group, the other had 30 students who belonged to the control group. Both groups were taught by the same instructor in a similar way; the difference was in the way both groups were assessed. While the experimental group was evaluated by using digital questionnaires on the website Quizizz, the control one was assessed in a traditional way using the same questionnaires but on paper.

The results showed that while both groups improved their grammar skills from the pre-test to the post-text, the experimental group got higher scores than the control group. According to Pham (2023) the reason could be that “elements such as feedback, points, a leaderboard, and attractive visual features can make the assignments more appealing to students” (p.8). Consequently, they ended up performing better. This research made a huge contribution to the present study, as an important part of the current one focused on the challenges that students struggled with in the classroom, as they complete activities as questionnaires, but in a digital

and physical way. This study was contemplated as a reference in the intended way to foster motivation and engagement in the population of the present study.

The ninth and last research was conducted in Turkey and was titled *The impact of gamification on secondary school students' grammar proficiency*, conducted by the researchers Koç and Soner in 2023. The aim of the research was to investigate the academic achievement of 6th grade students after being exposed to grammar learning in a gamified manner. The population consisted of 80 students enrolled in a private school in Turkey who were divided into two groups, one experimental and one for control, each one with 40 students respectively. In the experimental group the researchers sought to use the website Kahoot in class as the vehicle to teach target grammar while in the control group instruction of it was given in a traditional way without the use of gamification. During the whole research interviews were conducted with the students from the experimental group with the purpose of getting to know their perspectives towards learning grammar with gamification.

The results showed that the students who were part of the grammar instruction via Kahoot got better results in an achievement test and presented meaningful differences than the students who received the same content by book-based education. Regarding the pupils' perceptions about using Kahoot, overall, it was concluded that most of them had a positive vision of it, as they affirmed that learning grammar became more enjoyable and they could get a better retention of information in this way. This research contributed to the present one by displaying the importance of exploring the point of view of the population involved in the project beyond only setting the spotlight on the measurable results.

## **Theoretical Framework**

The following section provides the concepts that under the light of theory will be considered for this research project. Firstly, phygital gamification, which is the primary focus

of this study, here a definition of gamification as well as the component elements from it is presented. Second, motivation and engagement, as they composed one of the problems evidenced during the characterization of the population. Thirdly, grammar and vocabulary learning, since these were the parts of language intended to be fostered, bearing in mind that the population showed big difficulties while using them during the characterization stage.

### **Phygital Gamification**

The mere concept of gamification has evolved over the years with the constant advances in the technological field and with the implementation of digital devices into educational settings. One of the most notable definitions for gamification is provided by Deterding et al. (2011) who define gamification as “the use of game design elements in non-game contexts” (p. 10) which leads to transforming the user’s experience and elicit engagement. According to these authors, gamification lies in the Latin term of *ludus*, which is translated to English as *gaming* and bears the concept of being involved in an environment where rules, points systems, competence and striving against others for achieving certain goals or outcomes acquires meaning.

In this way gamification goes beyond using games in given settings but making the settings themselves become a game. In gamification design certain elements and mechanics, proper from games, must be adapted for the creation of the environment. These elements can vary depending on the purposes of the gamification designer. Nonetheless, gamifications may include components such as *leaderboards*, *awards*, *avatars*, *challenges* and *competition*. (Schöber, Janson, and Söllner, 2019).

In the first place, *leaderboards* are understood by Schöbel, et al (2019) as lists of users or players, where each participant has a score, in shape of points or other goods, that are displayed in a way that promotes competition Secondly, most gamifications make use of points



systems known as *rewards*. These are the goods added to a user's account for completing certain tasks or following an intended behavior. *Rewards* can be given in the shape of achievements or penalties when the outcomes are negative. (Thiebes, Lins, 2014). Thirdly, following Werbach and Hunter (2012 as cited in Schöber, et al. 2019) the *avatars* are “the visual representation of a player's character” (p. 645). They can be illustrations of an object, animal, or anthropomorphic being that represents the player or person who exists in real life in the outer world. Fourthly, the *challenges* are tasks that the users develop towards a goal.

Having the previous information in mind, users are rewarded after successful completion of challenges (Thiebes, Lins, 2014). Also, a central aspect in any gamification is *competition*, which is the condition of striving to overcome other individuals to achieve something which denotes a grade of superiority. Sometimes *competition* is not given under the striving to overcome others but overcoming other factors such as self-capacity and time. For instance, timers are often included to force users to undertake a task in a limited lapse of time, making them compete against it.

Now, when addressing Phygital Gamification, it is necessary to understand first the concept of *Phygital*. According to the Capacity Building Commission (CBT) the term *Phygital* is a combination of the terms *Physical* and *Digital*. It implies the use of digital tools along physical interaction elements in a blended experience for the user (CBT, 2023). This concept emerged initially from enterprise marketing and recently has become part of E-Sports. Within the educational field, the term has not been explored in depth yet to establish a theoretical basis.

Nonetheless, under the light of the tenets of the term phygital and gamification, it could be alluded to phygital gamification as the implementation of the game design elements in digital and physical representations in a simultaneous way. According to Goretti, et al. (2020), phygital gamification permits the use of digital stimuli with a physical and therefore material

part in the learning process for students Bearing this in mind, Phygital Gamification may display gamification elements such as competition, challenges, rewards, avatars, etc., on a screen as well as in a tangible good. For instance, a class leaderboard can be displayed on a screen, while the points or rewards that set it up can be physical badges obtained from the activities delivered by the students in the classroom. This new approach changes the perspective of common gamification as the classroom becomes an immersive space where the students participate.

### **Motivation and Engagement**

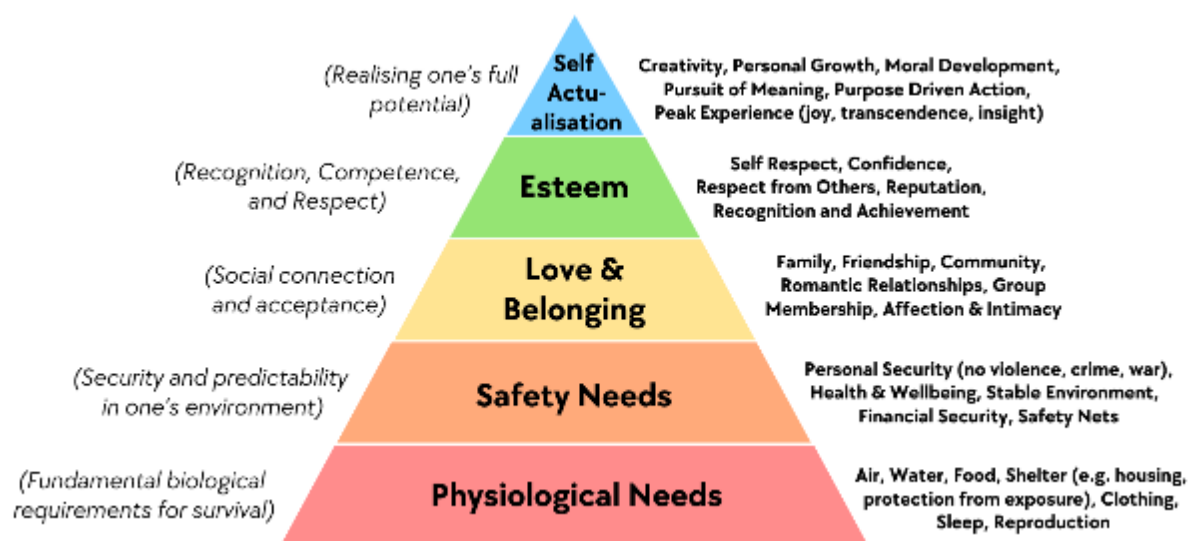
Even when these two concepts may appear to refer to the same meaning and their deep and undeniable relation can be a cause of confusion, they do not stand for the same. Thus, it is crucial to determine their differences for a better understanding of this research. According to Pinder (1998) motivation is the combination of internal and external forces that have a repercussion in the work-related behavior, and shape its form, positionality, magnitude, and duration. In other words, motivation refers to those psychic aspects that play a role influencing people to act in the desired way.

Similarly, according to Reeve (2009) motivation is understood essentially as those internal processes that energize and empower the reasons that an individual has to direct their behavior in a determined way. In those internal processes the individual does a balance between what Reeve calls *intrinsic and extrinsic motivation*. The first is led by three main elements: autonomy, competence, and affinity, which ultimately aim to fulfill the desire of psychological satisfaction. On the other hand, Reeve affirms that not all motivation comes from the inside even when that is the place where the processes start. The author proposes the existence of an extrinsic motivation where all the elements, pressures, expectations, among others, that the environment provides play a role for the individual to accomplish the intended goal. As he describes it is a sort of behavioral deal from the type “do this and you will get this” (p,84),

working as an incentive that creates in the individuals the imaginary of either getting attractive consequences or avoiding negative ones as a result of their actions.

Following this perspective, it is a matter of common sense that for having reasons to be motivated towards doing something, there must be a need or a motivating factor that leads the person to do so. These aspects can be tracked back in the theory proposed by Maslow in 1943 where he argued that motivation was the result of people attempting to achieve something that they did not have yet. This theory is exemplified in the shape of a pyramid with five levels of needs that go from the lowest and most basic needs to the most transcendental ones that must be satisfied in order to create motivation. Consequently, a person that has not fulfilled the basic levels of need in this hierarchy is susceptible to not having reasons to create motivation.

*Figure 5: Maslow's hierarchy of needs pyramid*

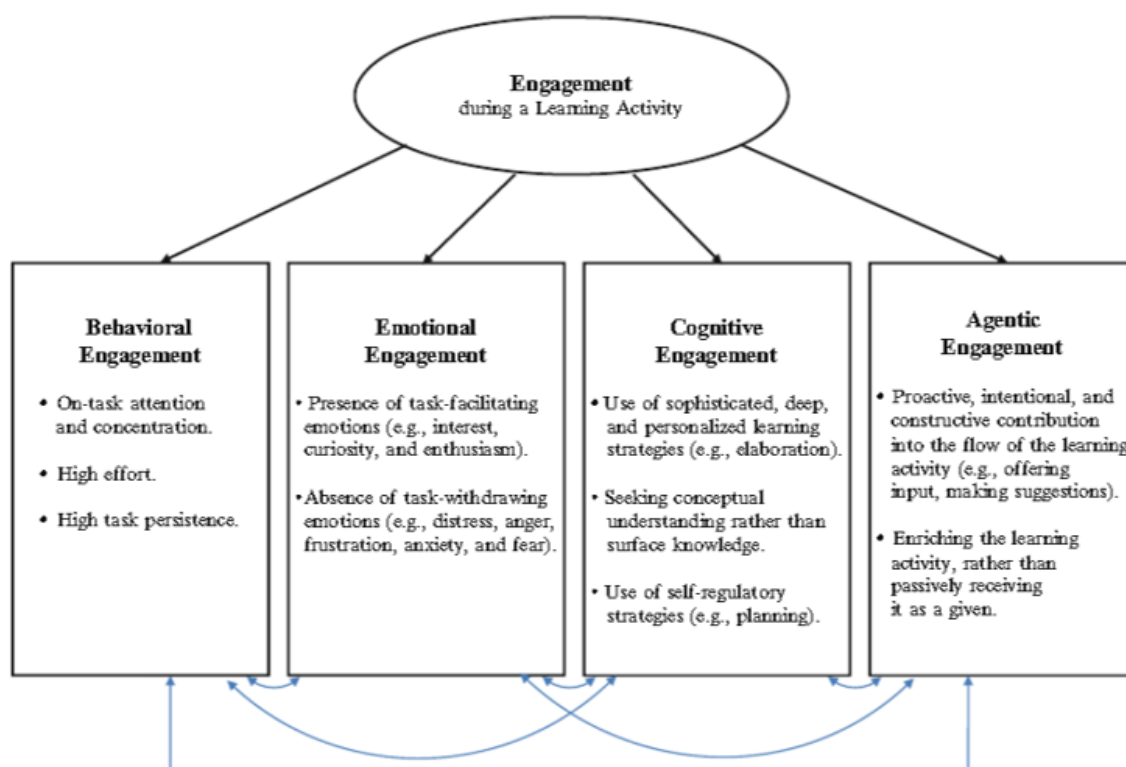


*Note: Adapted from A Theory of Human Motivation, by A. H. Maslow, 1943, Psychological Review, 50(4), pp. 370–396.*

Something important to consider is that while motivation is unobservable by itself, as it is part of the private psyche of the person, being denominated “a neural and biological process” (Reeve, 2012, p. 151), engagement is a physically observable behavior that is caused

by motivation. Moreover, Reeve (2012) establishes engagement as a multidimensional process that is perceptible through an individual's conduct, emotion, cognition, and agency. The first engagement dimension corresponds to the behavioral effort in terms of attention and persistence that individuals display to their task involvement. Within the emotional sphere, different states may appear such as interest, enthusiasm, and enjoyment. In the same line, some others known as “task-withdrawing emotions” (p.151) can also disappear, such as anxiety and frustration. Finally, the cognitive and agentic dimensions of engagement involve the individuals' mental efforts to reach deep understanding of the goal instead of only keeping what they are exposed to as the whole. Here their proactive contribution to the learning process is also assessed by taking an active and self-directed role. These four dimensions have bidirectionality relationships; this means that when one of them is triggered, it can simultaneously activate or reinforce the others.

*Figure 6: Reeve's four interrelated aspects of student engagement*



By understanding these four dimensions, engagement acquires a more complex connotation as this phenomenon transcends the mere act of paying attention to something and recognizes different viewpoints from which engagement can be tracked. Fostering this process within the class environment is significant, as Kearsley and Schneiderman (1998) state, when students are engaged, they feel close to the proposed activities participating actively in them, interacting with others, and taking an authentic disposition to class, which ends up driving their learning process in a more pleasant way.

### **Grammar Learning**

Considering the needs and the EFL knowledge of the population involved in the development of the current project, grammar learning was taken as a key factor for the research. According to Thornbury (2001) grammar learning is defined as the study of the formal language structures and the rules that dictate how sentences are formed in it. This author suggests that in synthesis, grammar's aim of study is syntax, which corresponds to the established word order in sentences, and morphology, understood as the study of how words are formed. This perspective emphasizes the issue that for a language learner "the ability both to recognize and to produce well-formed sentences is an essential part of learning a second language" (p.3)

Since the 20th century, grammar has been a controversial matter in language teaching, as many linguists criticize the issue of systematizing vast and complex corpus of knowledge, as languages are, being arranged in arbitrary rows and boxes. Authors like Swan (2002) even establish a difference between grammar learning and language learning, as he argues that learners may have an illusion of knowing a language after acknowledging the main rules of it and approving tests, but when they are forced to use it in practical ways, they lack vital elements.

In contrast to this, other perspectives, such as Richards and Renandya (2002) recognize the cornerstone role grammar has in language learning and argue that “without a good knowledge of grammar, learners’ language development would be severely constrained” (p. 145). Ultimately, these authors suggest that the discussion should not be about whether teaching grammar or not, but about recognizing the grammar elements learners need the most and how it can be contextualized in a real use scenario.

### **Vocabulary Learning**

The fourth construct used in the research is vocabulary learning defined “as the knowledge of words and word meanings” (Alizadeh, 2016, p. 22). Such knowledge must be understood inside the grammar perspective, as “grammar and vocabulary are not two separate components of a language” (Halliday & Matthiessen, 2014, p. 7). In this sense, Vocabulary learning encompasses a series of properties that go beyond the linguistic field, such as grammatical use, written form (spelling), speaking form (pronunciation), register and association. (Schmitt, 2000, cited in Alizadeh, 2016). Similarly, as Vygotsky would say in his work *Thought and Language*, (1986) “A word is a microcosm of human consciousness” (p.255), which means that words entail more than simple linguistic and representative structures to denote concepts, but they are the reflection of long evolution processes that languages have played within human communities, involving their way of thinking.

In line with Thornbury’s theory (2002) when learning another language, learners develop different psychological processes than when they learn words from their native language, as they have already created conceptual networks of associations between words (signifiers), their characteristics, and of course the mental referents they represent (signified). Hence, creating a second mental lexicon involves expanding the conceptual networks to create new vocabulary from the referents that are already installed in our minds. For instance, a native English speaker does not have to relearn the concept of *hand* when learning another language

but just expand its conceptual networks to associate the same concept with other signifier or word, for example, *mano* in Spanish.

### **Chapter III: Methodology**

This third chapter pretends to frame the type of research that was conducted, including the research design as well as the paradigm. Then the instruments used through the study for data collection are presented. Finally, the ethical considerations applied for this research are exposed.

#### **Research paradigm**

The paradigm followed in this study was qualitative, as this research looked to gather a cluster of information related to the unique characteristics that the people studied hold, which according to Cerda (1991) allows the researcher to understand the human behavior within a specific setting and the possible reasons that originate it. Overall, qualitative studies aim to consider the point of view of the participants in the research, as the researcher looks through the eyes of the participants, involving empathetic relationships towards them, and immersing themselves in the contexts where they participate (Mella, 1998).

On the other hand, qualitative research has a procedural nature, which conceives life as a succession of events that are interconnected and in constant change. (Aravena, M. et al. (2006). This part is crucial as it outlines how the participants' behavior is the consequence not only of what they do within the specific setting where they are studied, but also over all the surrounding conditions that their lives imply. To achieve this purpose of getting familiar with the dynamics of the group studied, five instruments of data collection were used during the research: observation, field diaries, questionnaires, a semi-structured interview and artifacts.

## **Type of research**

Alternatively, for the present study an action-research design was used, as it focused on describing the actions and behaviors of a group of seventh grade preteenagers of basic secondary school, to identify possible problems that could be addressed by the implementation of a pedagogical intervention after a process of reflection. In this sense, action research has been defined as a process where researchers evaluate and study problems with the aim to change them by “a small-scale intervention in the functioning of the real world and a close examination of the effects of such an intervention” (Cohen and Manion, 1994, as cited in Cohen, Manion, Morrison, 2002, p.297).

Bearing this in mind, there are some steps to consider when conducting action research. According to McNiff (2002) the first thing that the researcher has to do is to reflect on the effectiveness of the current practices carried out. Then, it is necessary to identify the aspects that could be improved. After that, the person in charge of the research has the responsibility to plan a path that seeks to improve such issues identified and put it into practice monitoring constantly the outcomes of its intervention. If needed, the plan can be modified during the process to tackle the situation appropriately. When the intervention is done, the researcher makes a process of evaluation and reflection regarding the strategy used.



Figure 7: Action research cycle



Note: Adapted from *Action Research: Principles and Practice* by J. McNiff, 2013. Diagram designed by the researcher.

## Instruments for data collection

### Observation

The first instrument implemented during the research was the observation, which is an instrument that enables researchers to study “the practices or doing displayed by social characters in *natural scenarios* during daily life situations, which are neither object of attention nor reflection by these social characters” (Jociles, 2017, p. 4). When applying observations, researchers are intended to understand the situations that they are describing and detect repeated patterns that constitute the natural dynamics of the object of study (Patton, 1990 as cited in Cohen, Manion, Morrison, 2002).

This author also suggests that there are three types of observation: structured, unstructured, and semi structured. Within the first one the researcher knows in advance what

is going to be observed or has some hypothesis that pretends to confirm. In the second, the opposite effect occurs, since the researcher does not have a clear idea of what is going to be observed, therefore the interpretations emerge during the observations. In the third, an agenda of what is going to be observed is established although the aspects observed cannot be predicted in a systematic way.

Therefore, for this research the third type of observation was used, as there were some established elements to be observed during the pedagogical intervention. These elements were the relationship that phygital gamification had with students' engagement and motivation, and the relationship that it had with learning English as a foreign language through grammar and vocabulary learning.

### **Field diaries**

On the other hand, field diaries were implemented in this study (see annex 3) as the instrument that allowed the researcher to record the observed events (Cerdeja, 1991). When using field diaries, the researcher transcribes everything happening in the context of the study into a paper or a digital note. Additionally, in line with Cerdeja (1991) it is important to make the precision that the information registered "must be objective and its data precise" (p. 253), which means that the description done is reliable as it avoids the point of view of the observers.

### **Questionnaires**

The questionnaire is a data collection instrument based on a series of structured questions that can be administered without the presence of the researcher (Cohen, Manion, Morrison, 2002). These authors suggest that when designing questionnaires some steps must be taken to avoid an eventual unappropriated use of the instrument, such steps are: First, decide the objectives of the questionnaire, as well as the sample where it will be used. Second, create topics relevant to the sample and the data desired to be addressed in the research. Third, create

a series of questions with these items. Fourth, pilot the questionnaire before applying it to redefine aspects that do not fit well. And finally, administering the final questionnaire. In this research, at the end of each cycle a questionnaire was implemented in order to know the student's perception about the phygital gamification and pedagogical intervention that was being applied in the classroom (see annex 4).

### **Semi Structured interview**

Semi Structured interviews are qualitative data collection instruments whose basis lies in delimited open questions that are tied to specific themes of analysis. Here, the interviewer engages with a participant subject to explore emerging topics, generating rich and in-depth data. What characterizes this type of interview from others is the feature that questions are not constrained to follow a strict sequence. As this tool enables the researcher to use open questions, it creates flexibility in regards of the opinions expressed by the interviewed and the nuance of their answers.

Additionally, the researcher has the possibility to modify the questions during the lapse of the interview in favor of the needs of the research and according to the data that the interviewee offers. In this type of interview, the aim is to understand the complex behavior of the members of a society without imposing a prior categorization that may limit the field of research. This type of interview was conducted with the head teacher with the aim of knowing her perceptions and analysis of the whole intervention, as she was present during the whole process (see annex 5).

### **Student artifacts**

Evangelou. et al. (2019) define artifacts as “any human-made object” (p. 2) that conveys meaning in form and function. In the school environment, artifacts are any object created by the students to achieve a learning goal, these ones vary depending on the teaching

purpose and among them there can be written outcomes, handicrafts, drawings, and objects molded with clay (Delgado and Chapeton, 2015). For the present research, worksheets, scripts, fill-in-the-blank exercises, word searches, and a short films were considered students' artifacts to collect qualitative data about their performance during the intervention (see annex 6).

### **Ethical considerations**

It is relevant for this study to mention that all the information gathered with the different instruments used followed some procedures to protect the integrity of the participants involved in the research as well as the confidentiality of their data. In the first place, as the research was conducted with an under-eighteen population, an informed consent (see Annex 7) was delivered to the parents of the students, so that they could be aware about the processes that were going to be performed with their children during the lapse of the research. Then, the names of the students were hidden from the instruments used such as field diaries, and the questionnaire, being replaced by superficial mentions such as a male/female student, to protect their anonymity.

## **Chapter IV: Pedagogical intervention**

This chapter presents the pedagogical approach used for the instruction as well as the concerning visions of curriculum, EFL classroom, and language. At the end, the instructional design to carry out the gamified proposal as the pedagogical intervention is presented.

### **Pedagogical Approach**

This study aimed to conceive collaborative learning as the way in which students can achieve learning goals by working collaboratively with their peers to complete a given task (Bruffee, 1999). With collaborative learning students take an active role in their learning process, since they are expected to share their ideas, get to know other points of view, analyze

problems and find solutions towards them, which leads to an interactive way to learn, apply and conceive knowledge within the class environment (Ramos, Cattaneo, de Jong, Espadeiro, 2022).

There are some aspects to be considered when using collaborative learning. First, although the focus is on the overall participation that groups of students do towards a task, it is important to underline that each individual takes action making meaningful contributions and encouraging one another. Second, the teacher takes an active role too, since it is the one who plans the tasks for group interaction, monitors the activities and provides feedback towards the students' outcomes (Ojie-Ahamiojie, 2024).

### **Vision of curriculum**

The concept of *curriculum* is certainly polysemic, since its meaning varies depending on the authors and the historical background it is being referred to. Nonetheless, there are theoretical foundations that prevail in time to underline a universal meaning of the curriculum. Considering this, the vision of curriculum for this study lied on the theory of Casarini (2012) who citing (Neagley and Evans, 1967) defines curriculum as “the set of planned experiences provided by the school to direct students to achieve the established learning goals according to their abilities” (p. 5). Which indicates that a curriculum is a plan of intervention designed by the members of a school where different processes such as planning, development of contents, and evaluation are involved, to guarantee a certain level of learning in students after a teaching process (Casarini, 2012).

### **Vision of EFL classroom**

In regard to the vision of the EFL classroom over this study. It was conceived as the place that offers an interactive experience, where students participate in the progression of the

class, rather than being forced to take a *student role* in a dominated lecture approach (Shpancer, 2004). Traditionally, both students and teachers are intended to follow prototypical roles during the class of English. In the case of the *students' role*, it lies on passivity, copying notes, and remaining sitting in silence while the teacher follows its prototypical *teacher role* of conducting lectures with their speech and by following the principles of a textbook. According to Shpancer (2004), this way of approaching a class not only limits group interaction, but also makes the class boring for students, since they are only having a constant load of input without any practical reflection.

Therefore, the EFL classroom is expected to be the place where these patterns are not favored, and where through the constant interaction through class performance, students can take a different attitude towards their learning process of the foreign language. In the case of this project, it is intended to direct this student interaction using phygital tools to create an interactive environment where students are challenged to participate in class and undertake different tasks while they learn new vocabulary and grammar.

### **Vision of language**

When referring to language learning, grammar and structures are often understood as central components. Much discussion has been developed around this issue, as some authors, such as Halliday (1978), suggest that languages have a social dimension in which meaning is co-constructed through interaction and therefore approaching language solely as a set of linguistic rules may be insufficient to fully understand language learning.

Bearing this in mind, in this study grammatical knowledge is perceived as an essential foundation for language development, because it provides learners with the structural basis necessary to organize meaning and communicate when they do not have the conditions to learn the target language in a natural environment through constant exposure. To avoid the

limitations caused by the abstract nature of grammar, it must be integrated with language use in context, giving importance to the role it has for communication and social interaction.

As a result, learning a language is not just accumulation of isolated knowledge, but also the use of it as languages are the means through which social interaction is promoted (Halliday, 1979). Thus, a language is understood only within a specific context through its use, when people write and speak, they are manifesting a unique perspective that language allows them to create.

### **Pedagogical intervention**

The pedagogical intervention consisted of three cycles that incorporated elements of collaborative learning. Each lesson in the three cycles intended to improve engagement and motivation in class while students worked on collaborative tasks aimed at learning vocabulary and fundamental grammar topics necessary to accomplish the final task of creating a news show. All this was developed by using phygital gamification as a pedagogical tool, where different digital and physical tools were used to conduct the class, creating a gaming environment over the 12 sessions. In the following part, there are descriptions of the digital tools ClassDojo and Genially (see annex 8), as well as the physical tools (see annex 9) that under the light of theory, enabled the teacher in training to create the elements that composed the gaming environment.

### **Digital Tools**

#### ***ClassDojo***

At first, the platform ClassDojo was used as the basis of the gaming class environment for this project. ClassDojo is an online free-use behavioral management platform used mainly by parents and teachers, which offers the possibility to give feedback to certain tasks and behaviors from students in a digital way. In the case of this project, the platform gave space to

the creation of a digital classroom where some of the key aspects of the class were controlled. First each student had an avatar on the platform that represented them in the class. Second, the platform was the place where the points in the shape of diamonds were accumulated and displayed to the students. Third, the tool controlled other aspects such as taking control of time for activities, and creation of work groups.

**Digital avatars.** Each student had a personal avatar that represented them in the class. These avatars were selected taking into consideration the age of the students and the interests they mentioned in a questionnaire. This led to having characters from series such as Dragon Ball, Pokémon, Powerpuff girls, SpongeBob, and games such as Sonic, Mario, Minecraft, among others.

**Digital scoreboard.** The digital avatars appeared on a screen every class with the respective diamonds that the students had obtained in the last class. This enabled them to know how their performance was in the previous class in order to improve or keep a good performance in the current class. This manner of competition was not given under the comparison with their peers but with themselves.

**Digital awards.** The stimuli of the classes focused on points obtained in the shape of diamonds which later turned into numerical grades. These diamonds were obtained in each class after completion of certain activities proposed by the teacher, such as class participation, completing the challenges of the game Word Warrior, presenting an outcome activity, and homework. The diamonds were displayed on the screen with the platform ClassDojo. On the other hand, there were negative diamonds, which students obtained when they showed a behavior of indiscipline, such as being rude to peers or to the teacher, copying from others, and not following the instructions indicated by the teacher.



These negative awards could subtract one of the obtained diamonds. The maximum number of diamonds per class was 10 and the grade was assigned dividing the number of diamonds into two. For example, a student who had 10 diamonds at the end of the class, had a performance grade of 5,0, while a student who only obtained 5 diamonds would have a performance grade of 2,5. With this in mind, the students were intended to obtain a determined number of diamonds per activity that later turned into a numerical grade.

### ***Genially***

The platform Genially was used as a second cornerstone of this gamification. Genially is an online free-use platform that allows its users to create, design and share all kinds of visual and interactive content using templates already created by the platform and other users. It enabled the teacher to create a learning game titled Word Warrior. The game consisted on a warrior who traveled across galactic islands defeating villains while some of their language challenges were completed. The game had 10 levels, and on each level a grammar topic was approached through diverse challenges that the students completed.

**Digital challenges.** The challenges of the game were approached during each level and they intended to make students put into practice the grammar and lexicon information they had learned during the class time. Among the diverse challenges undertaken in the intervention, fill-in the blanks exercises were considered, where students completed real life sentences by choosing an answer option or conjugating one verb. Another one was word searches, where students were intended to find vocabulary as it appeared or as the teacher asked, for instance a student could be asked to find a verb conjugated in past in the word search when in the list it appeared on its base form, crosswords, match the words, false or correct, and find the differences on an image. These challenges of the game were shown on the tv screen and were solved with a physical student control guide.

## **Physical tools**

**Physical avatars:** Physical avatars were created in the form of crafted cardboard badges that were delivered to the students each class. These badges had the image of the digital character, as well as the name of the student. This physical element aimed to improve some tasks for the teacher such as taking attendance and giving participation. They also intended to give a sense of belonging to the students within the class.

**Physical awards.** For class participation physical pieces of printed diamonds were given to the students, who at the end of the class redeemed them on the digital platform. The use of these physical diamonds allowed the students to get involved in the gaming environment, ever since they could have something to touch and use. On the other hand, just as in the digital way, there was a negative award when a student showed bad behavior during the class. Students were shown a red card which meant that a diamond was going to be subtracted from their account.

**Physical challenges.** The students were the ones in control of making the decisions to overcome the challenges of the game Word Warrior. For this, a paper with the same challenges that appeared on the screen was delivered to the students in class time. They had to solve the challenges either in pairs or individually when they were projected one by one on the screen. Each challenge had a time limit depending on its variation, 30 seconds for fill-in the blanks and false vs correct exercises, 2 minutes for matching the words, and 5 minutes for word searches, crosswords, and find the differences exercises.

The students had to complete these challenges in the given time and stand up from their chairs to throw the control guides into a color box depending on the answer they had chosen. If they did not complete the challenge during this time, their worksheet was not accepted, and they had to wait for the next challenge. There was one box for each answer that

appeared on the screen, Red Box, Blue Box, Yellow Box. After this was done, the teacher showed the correct answer to the challenge and graded the ones who had answered correctly with check marks by revising the papers of the box with the correct answer. This way, the challenges were conducted over the sessions and at the end the teacher assigned a number of diamonds depending on the checkmarks that appeared on the paper of each student. This strategy gave place to include kinesthesia into the gamification and facilitated the assessment process, since the teacher graded only the papers located inside the correct box, reducing the time for this task.

Following the approach chosen for the implementation, three different cycles were created to solve the problem identified using phygital gamification. Through the first two cycles, students were intended to build a grammar foundation and broaden their vocabulary through active participation and collaborative work which supported the development of a final project based on the creation of a news show in the third cycle. This project was selected to provide students with the opportunity to use the target language in an authentic and meaningful way, going beyond artificial tasks and allowing them to perform communicative skills while they displayed their own reality and context. The three cycles were distributed as follows:

### **Cycle 1 Building a basis (pre-task stage)**

In the first cycle of the intervention, the objective was to provide learners with the vocabulary and grammar structures necessary to create short sentences in present and past, with the intention to offer them sources to create a news show where they could use English in an authentic way. The first 6 lessons focused on the input of present simple, past simple, present progressive and past progressive. These topics were delivered to students using digital presentations, videos, flashcards, papers, and written information on the whiteboard. After this input was given to the students, they practiced it with the tool Word Warriors to give them the

opportunity to recall the information just seen. Finally, once the game was over, the students had to deliver an activity with an outcome, where they used the topics in a contextualized way.

*Table 2: Scheme of cycle 1 of the intervention*

| SESSION | TOPICS                                                                                                                                                              | VOCABULARY / GRAMMAR                                                                                                                                                    | OBJECTIVES                                                                                                                                                                              | OUTCOMES                                                                                                                                                                 |
|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1       | <ul style="list-style-type: none"> <li>Routines</li> <li>Seasons of the year.</li> </ul>                                                                            | <ul style="list-style-type: none"> <li>Present simple tense for affirmative, negative and interrogative sentences.</li> <li>Weather and seasons in the world</li> </ul> | <ul style="list-style-type: none"> <li>To recognize the structure of present simple tense.</li> <li>To identify vocabulary associated with weather and seasons.</li> </ul>              | <ul style="list-style-type: none"> <li>Oral exercises using present simple tense (one by one)</li> </ul>                                                                 |
| 2       | <ul style="list-style-type: none"> <li>Common verbs for routines</li> <li>What do I do in Bogota?</li> </ul>                                                        | <ul style="list-style-type: none"> <li>Present simple tense for affirmative, negative and interrogative sentences.</li> <li>Places and objects in the city.</li> </ul>  | <ul style="list-style-type: none"> <li>To recognize the structure of present simple tense.</li> <li>To identify vocabulary associated with everyday life actions in the city</li> </ul> | <ul style="list-style-type: none"> <li>Exercises paper using present simple tense (whole group)</li> </ul>                                                               |
| 3       | <ul style="list-style-type: none"> <li>Describing my current and past routines</li> </ul>                                                                           | <ul style="list-style-type: none"> <li>Present simple vs Past simple</li> <li>Time expressions in present and past.</li> </ul>                                          | <ul style="list-style-type: none"> <li>To recognize the differences in use and structure of present simple and past simple.</li> </ul>                                                  | <ul style="list-style-type: none"> <li>Exercises paper using either present simple or past simple tense (whole group)</li> </ul>                                         |
| 4       | <ul style="list-style-type: none"> <li>27 most common adjectives turned into comparatives</li> </ul>                                                                | <ul style="list-style-type: none"> <li>Comparative adjectives</li> </ul>                                                                                                | <ul style="list-style-type: none"> <li>To identify the use of comparatives</li> <li>To induce the way they are created according to the number of syllables in a word.</li> </ul>       | <ul style="list-style-type: none"> <li>Exercises paper using comparatives with "er" or with "most" (whole group)</li> </ul>                                              |
| 5       | <ul style="list-style-type: none"> <li>Time expressions for routines and for actions in progress</li> </ul>                                                         | <ul style="list-style-type: none"> <li>Present simple vs Present progressive</li> </ul>                                                                                 | <ul style="list-style-type: none"> <li>To recognize the differences in use and structure of present simple and present progressive</li> </ul>                                           | <ul style="list-style-type: none"> <li>Exercises paper using either present simple or present continuous tense (whole group)</li> </ul>                                  |
| 6       | <ul style="list-style-type: none"> <li>Touristic review in an accommodation in Bogota</li> </ul>                                                                    | <ul style="list-style-type: none"> <li>Past simple vs Past progressive</li> <li>Regular and irregular verbs in past</li> </ul>                                          | <ul style="list-style-type: none"> <li>To recognize the differences in use and structure of past simple and past progressive</li> </ul>                                                 | <ul style="list-style-type: none"> <li>Written exercise scanning past verbs in a touristic review.</li> <li>Written exercise using past continuous sentences.</li> </ul> |
| 7       | <ul style="list-style-type: none"> <li>Common vocabulary in tourism.</li> </ul>                                                                                     | <ul style="list-style-type: none"> <li>Review of previous topics</li> </ul>                                                                                             | <ul style="list-style-type: none"> <li>To make a review of the four tenses studied in class: (Present simple, present continuous, past simple, past continuous.</li> </ul>              | <ul style="list-style-type: none"> <li>Written exercise looking for verbs in the classroom.</li> <li>Exercises paper, using the four tenses.</li> </ul>                  |
| 8       | <ul style="list-style-type: none"> <li>always, right now, yesterday, at the moment, never, last week, usually, today, one year ago, in 2010, on weekends</li> </ul> | <ul style="list-style-type: none"> <li>Time expressions in present and past review</li> </ul>                                                                           | <ul style="list-style-type: none"> <li>To recognize the different time expressions used according to the tenses in present and past</li> </ul>                                          | <ul style="list-style-type: none"> <li>Participation exercise matching expressions with the intended tenses</li> </ul>                                                   |

## Cycle 2 Constructing a news program (task stage)

In the second cycle of the implementation, this structure of input-game-task was maintained during the lessons, with the difference that the explanations and activities were oriented to the creation of the segments of the news show. The input given in the beginning of the lessons consisted of the macro-structure of the intended news show (local news, flashback segment, live report) and the micro-structure of each of these sections as well as a revision of previous topics (Present tenses and Past tenses).

The elements that the news show should include, as well as the criteria for this project, were presented to the students. The students were asked to think about funny, realistic and interesting news at home that could help them to the construction of this news show in class

with their peers. The news show was built by students during the three lessons established, where the teacher monitored the advancements, and assessed the outcomes obtained every single class.

*Table 3: Scheme of cycle 2 of the intervention*

| SESSION | TOPICS                                                                                                                                                         | VOCABULARY/GRAMMAR                                                                           | OBJECTIVES                                                                                                                                                                                     | OUTCOMES                                                                                                                                             |
|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9       | <ul style="list-style-type: none"> <li>Structure and resources to create an artificial news program</li> <li>Structure for the presentation section</li> </ul> | <ul style="list-style-type: none"> <li>Present simple</li> <li>Present continuous</li> </ul> | <ul style="list-style-type: none"> <li>To identify the structure for the news program (presentation, local news, flashback segment).</li> <li>To create the presentation section</li> </ul>    | <ul style="list-style-type: none"> <li>Groups created</li> <li>News programs' names</li> <li>Assigned roles</li> <li>Presentation's draft</li> </ul> |
| 10      | <ul style="list-style-type: none"> <li>Structure for the local news segment</li> </ul>                                                                         | <ul style="list-style-type: none"> <li>Present simple</li> <li>Present continuous</li> </ul> | <ul style="list-style-type: none"> <li>To review the present simple and present continuous in the first two segments of the news program.</li> <li>To create the local news segment</li> </ul> | <ul style="list-style-type: none"> <li>Presentation's final version</li> <li>Local news' final version</li> </ul>                                    |
| 11      | <ul style="list-style-type: none"> <li>Structure for the flashback news segment</li> </ul>                                                                     | <ul style="list-style-type: none"> <li>Past simple</li> </ul>                                | <ul style="list-style-type: none"> <li>To review the past simple in the third section of the news program</li> <li>To create the flashback segment</li> </ul>                                  | <ul style="list-style-type: none"> <li>Flashback segment's final version</li> </ul>                                                                  |

### Cycle 3 Showing my outcomes (post-task stage)

In the third cycle, there was no grammatical or lexical input. The game Word Warriors was played for the last time, to revise students' performance in all the topics seen after the whole intervention. Then, the students were intended to present their news show project in class, relying on a digital produced video that they had to create and edit. The task was assessed in five categories: Use of grammar studied in class, vocabulary, pronunciation, task accomplishment, and video quality.

*Table 4: Scheme of the cycle 3 of the intervention*

| SESSION | TOPICS                                                                 | VOCABULARY/GRAMMAR                                                                                                                                                                                                            | OBJECTIVES                                                                                                                | OUTCOMES                                                                                                                                                                 |
|---------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12      | <ul style="list-style-type: none"> <li>News program project</li> </ul> | <ul style="list-style-type: none"> <li>Present simple</li> <li>Present continuous</li> <li>Past simple</li> <li>Verbs in past</li> <li>Places in the school and city</li> <li>Time expressions in present and past</li> </ul> | <ul style="list-style-type: none"> <li>To present the video project in class created by each group of students</li> </ul> | <ul style="list-style-type: none"> <li>Class presentations</li> <li>Assessment: Use of grammar, vocabulary, pronunciation, task accomplishment, video quality</li> </ul> |

## **Chapter V: Data analysis and findings**

Throughout this chapter, the data analysis and the findings of the study are presented. In the first place, the procedures for data analysis are outlined, including the approach and perspective adopted in this research. Moreover, the categories of analysis are introduced together with relevant data samples that support the results obtained. Finally, a discussion of the findings is provided.

### **Data analysis procedure**

The data analysis was conducted following the principles of grounded theory, in which data is progressively integrated, and categories emerge from the analysis process. As explained by Creswell (2007), this approach aims to explain actions, interactions, or processes through the interrelation of categories derived from participants' data.

For this study, the data consisted of observations recorded in field notes, questionnaires, students' artifacts, and an interview with the head teacher. Regarding the field notes, these were created once the intervention sessions were finished to guarantee their fidelity. These included one column of description of the events that took place in the sessions and one column for the interpretation of these events. On the other hand, within the students' artifacts the paper guides that students used during the interventions with the phygital gamification are considered, as well as a final video project that students delivered. Finally, a structured interview was conducted with the head teacher to get to know her thoughts and perspectives regarding the process with the group, since she was present during the whole intervention.

For the data analysis process, the categories appeared following a process of coding, that in line with Braun and Clarke (2021) is a method where through codes and labels data can

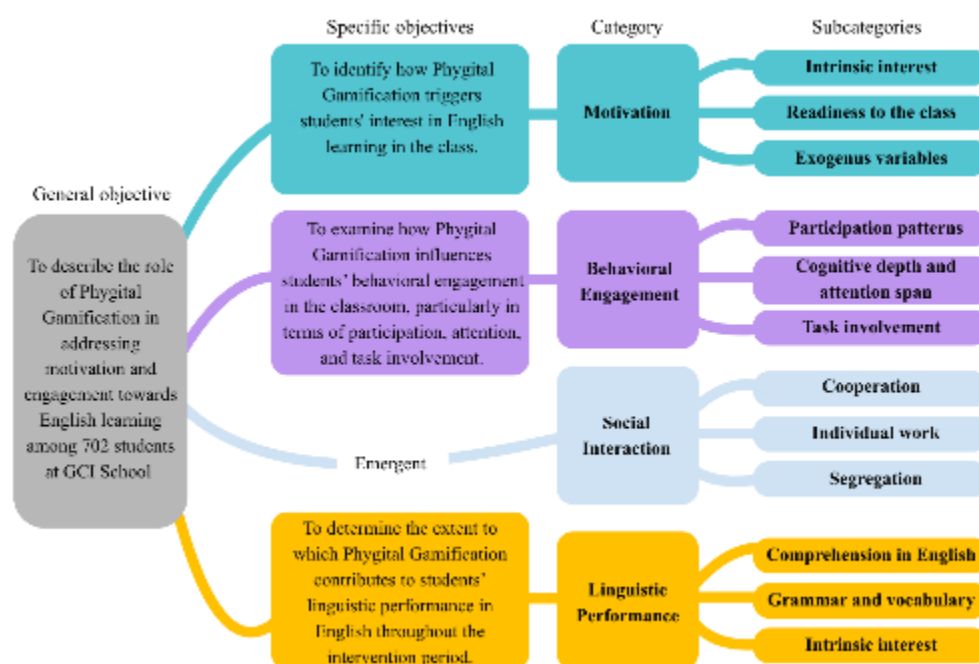
be systematically organized to find out the pieces of information that are potentially relevant to the research question.

Thus, codes emerge as a combination of two approaches; inductive, which means what comes directly from the data, and deductive, which refers to what appears in the process of analysis. As the authors suggest “we always bring something to the data when we analyze it, and we rarely completely ignore the data themselves when we code for a particular theoretical construct” (Braun and Clarke, 2021, p.3). Following this process, colors and labels allowed to establish the patterns that participants displayed the most during the whole intervention.

Concepts such as motivation, behavioral engagement, and linguistic performance, emerged as major categories with the deductive approach, since they were linked with the initial goals of the study. After this, another major category labeled as social interaction, and other 12 subcategories that complemented the major ones, emerged with an inductive process from the data collected. This enabled the exhibition of specific pervasive patterns observed during the intervention.

## Categories of analysis

Figure 8: Categories of the analysis



### Category 1. Motivation.

The first main category of the research was motivation, where other three subcategories emerged as part of the process of analysis: intrinsic interest, readiness to the class and exogenous variables. As was outlined in the theoretical framework, motivation is understood as an unobservable domain that is an exclusive part of the human psyche. Therefore, the pieces of information contemplated for this category are mainly made of perceptions from the students, from the teacher in training, and from the head teacher. Such perceptions were related to the purposes and reasons of each one to learn English, attend to class, and follow the dynamics that were proposed in it. External situations that might affect students' disposition to learn were also considered.

#### Category 1.1. Intrinsic interest.

In this category, the reasons students expressed to learn English or to attend the English class based on their life goals and their understanding of the importance of the language in the



world are presented. These social perspectives are crucial to direct each one's behavior, since as Reeve (2012) points out, the conditions of the environment can either nourish or decrease the motivation that a person has to do something. Following this line, students who are raised in environments where English is not perceived as essential for success in life due to the limited cultural presence it has in the context, lack of economic opportunities, and absence of personal aspirations to learn it such as international travel, are more likely to participate in class only to accomplish the academic requirement leaving aside a genuine interest for the language. This tendency is supported by the head teacher's perspective who knows at first hand the socio-cultural context where the students come from and recognizes the influence that it has on their motivation for English:

*“When a student understands that their education and effort will take them to a successful future, where there is personal and economic stability as well as social recognition, they may invest effort towards that goal. However, if they observe that their parents did not complete primary or secondary school and they currently work in informal or independent jobs where money is obtained immediately, they tend to believe that this is the most effective way to have success”. (Interview with the head teacher, own translation).*

This perspective suggests that motivation of the students to learn English also comes from their socio-cultural background as their families and the environment where they grow, assign the language a non-relevant value as it is not directly linked with the idea of success and future opportunities in their immediate context.

On the other hand, it is plausible to assume that some students initially attended classes without intrinsic motivation; however, throughout their learning process, they developed a sense of purpose for participating as a result of the dynamics that happened within the learning

environment. In this regard, it was found that the perception of the students towards the class of English was positive after the intervention, since they found that it was possible to learn English and have fun at the same time. Therefore, for some students the classroom evolved from being a mandatory requirement into a space where they felt comfortable and engaged, as seen in the next example:

1) Describe con tus propias palabras qué fue lo que más te gustó de las actividades del profe (por ejemplo: los diamantes, el avatar, los desafíos de responder la guía y moverse a las cajas, el proyecto del noticiero).

He parecido muy chaveros y divertidos y pues a uno le  
daban como más ganas de estar en las clases por las  
actividades. y me parecían mas chaveros el de las cajas

*(Questionnaire about students' final thoughts on the intervention, student 22)*

At first look, this scenario may appear ideal, as students who initially had a lack of motivation to attend class found a reason to engage. However, this phenomenon requires analysis from different perspectives. This consideration appears from the findings, which indicate that several students developed a sort of focus on gamification elements and classroom activities, to the point that their attention got distance from the intended learning objectives. This situation can be interpreted through the term *over justification effect*, as described by R. Lepper, Greene and Nisbett (1973), who argue that the introduction of extrinsic rewards to motivate behavior may, under certain conditions, disrupt intrinsic interest in the activity itself.

1) Describe con tus propias palabras qué fue lo que más te gustó de las actividades del profe (por ejemplo: los diamantes, el avatar, los desafíos de responder la guía y moverse a las cajas, el proyecto del noticiero).

lo que mas me gusto de las clases del  
profe fue cuando nos daba diamantes

*(Questionnaire about students' final thoughts on the intervention, student 11)*

In this sense, this means that students may have changed their focus of learning the language as a purpose to get engaged with the system and the dynamics proposed to learn the language. Consequently, this situation decreased their intrinsic motivation, making them more dependent on the teaching method. This phenomenon was also observed by the head teacher who expressed that

*“At times it was just competing for competing, without really being aware of what they were learning or paying attention to the details.” (Teacher interview, own translation).*

In relation to this, the authors mentioned before, suggest that fostering spontaneous and self-driven interest in learning should be prioritized over praxes focused on external incentives.

### **Category 1.2. Readiness for the class.**

On the other hand, Reeve (2009) argues that internal motivational processes play a crucial role in directing behavior and energizing individuals toward goal-oriented action. In relation to this, the findings revealed that, overall, students demonstrated increased confidence, which may be related to the sense of comfort that they developed in the classroom environment. As students began to perceive the classroom as their own space, they felt more comfortable making mistakes without fear of being judged, which at the same time supported their gradual learning process, as the head teacher described:

*“A positive aspect is simply engaging students, building their confidence, and helping them make their English learning more effective. It is important that they understand the purpose of communicating in English. Confidence plays a central role, especially in terms of self-esteem, by showing them that they are capable—for example, telling them: ‘Look, you can do it; you are earning points; you are achieving it.’ (Teacher interview, own translation)*

This perspective suggests that transforming the classroom into an environment where confidence and support are promoted has an influencing role in enhancing students' readiness to participate in class. On this basis, students began to feel capable and more willing to take risks without fear of being judged, and therefore they acquired a different disposition to the language and the class, as this fragment suggests:

5) ¿Crees que estas actividades mejoraron o empeoraron tu concentración y comportamiento en clase? Da un ejemplo  
mejoraron porque siempre que pasaba a  
frente a exponer me ponía nervioso pero ya  
no

(Questionnaire about students' final thoughts on the intervention, student 30)

In pursuit of this goal the teacher plays an important role in making students feel at ease and reducing anxiety in the classroom. By showing support to students and reinforcing the idea that their contributions to the class are not only valid but useful, they begin to abandon withdrawal or avoidance patterns such as anxiety and pressure and become more involved in the lesson. As the data suggests, participation increases when students perceive this support:

*"The teacher in training congratulated him and gave him a diamond of cardboard, which the student received enthusiastically. Then, the teacher in training asked the whole group, (...) This time over 6 students raised their hands"* (Field note 4, lines 76–82).

### **Category 1.3. Exogenous variables.**

Beyond the internal factors that shaped students' attitudes toward learning English, this category also considers the influence that the external variables present in the educational environment had on students' willingness during class time. During the intervention, it was seen that the class was a constant object of interruptions which deviated the attention of

students and affected the natural pace of the class. This was described by the teacher with the following words:

*“We had lunch periods, extracurricular activities, or sometimes students would arrive late. These were factors that interrupted the flow of the class. When those factors are removed, there tends to be greater concentration.” (Teacher interview, own translation).*

As Reeve (2009) described, motivation also comes from the outside, since the surrounding environment can either support or obstruct the reasons that the individual creates internally to act as desired. In this sense, these unexpected disruptions observed may have caused limited students’ opportunities to remain focused and actively involved in the learning process. Moreover, these situations had a direct influence on the order of the class, as it became difficult for the teacher in charge to re-establish control of the class and student focus, affecting the effectiveness of the lesson, as described in the field diary #5:

*After 20 minutes of lunch, the students and the head teacher arrived at the classroom. Even when they had the opportunity during lunchtime to interact, they continued talking, and almost none of them paid attention to the teacher in training, who was trying to regain their attention in order to continue the class (Field note 5, lines 98–103).*

These institutional dynamics affected in a degree the continuity of classes and limited the emergence of optimal learning conditions in the classroom, which had a direct effect on students’ extrinsic motivation. These elements were valuable to consider, as if students did not recognize a serious commitment to the learning process in their environment and their peers, they did not perceive a reason to create their own commitment.

Overall, these findings revealed that students’ motivation to learn English and follow the methodologies of the teachers in class, was determined by both internal and external

conditions that were complex and that required deepness when analyzing the reasons why they behaved in a very specific way. The interaction between their socio-cultural background and their dynamics at school cannot be ignored, as it is in this space where motivation is constructed, expressed, and, in which it has more possibilities to be shaped.

## **Category 2. Behavioral engagement**

The second major category was behavioral engagement, understood in this research as the perceptible patterns of behavior that are a response of motivation (Reeve, 2012). In this research these patterns could be tracked from three indicators: first, participation which reflected the ways in which students actively contributed to the development of the class. Second, cognitive depth and attention span where it was possible to analyze how students processed information and kept their focus during the classes. And finally, task involvement where the emotional, behavioral, and cognitive investment by the students in the tasks proposed was shown.

### **Category 2.1. Participation patterns.**

Regarding student participation patterns, the findings indicate that they increased during the intervention, as a consequence of the use of external rewards. At the beginning, only a few students decided to participate voluntarily. However, once the strategies were included, more students became actively involved, as the field diaries reveal:

*“Some students began to participate and the teacher in training wrote their examples on the whiteboard and gave them one diamond every time they offered a proper example. (...) student participation increased as they started to understand the topic more and realized that participation was important to obtain diamonds.” (Field notes 5, lines 37–42).*

As it was outlined, participation grew as students perceived its value within the reward system and probably the class as well. When students react to their environment and decide to participate in it by their own initiative, they begin to be agentic engaged, as Reeve states: “students’ constructive contribution into the flow of the instruction they receive (...) students find ways to enrich the learning activity or to make it more personal, interesting, or need-satisfying” (2012, p. 162). Moreover, driven by the competence system, participation was developed in different ways, which as the head teacher suggests, made students be more attentive of their actions during the class:

*“So, they would look at their performance, which came from teamwork, but each of them also had their own points. So, they were competing not only as a group—with their friends—but with the rest of the class too and even against themselves (...) and that always kept them alert.” (Teacher interview, own translation).*

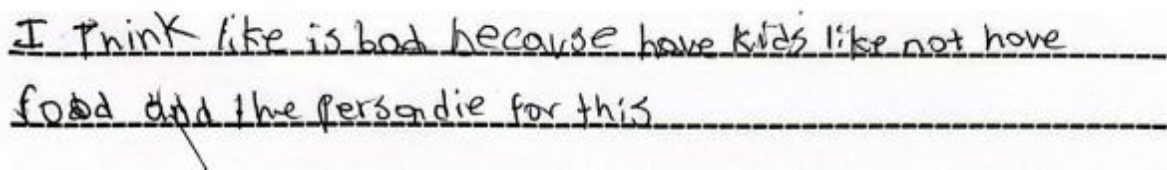
This illustrates a transition from a classroom where participation was limited, and most of the cases promoted by the teacher, to a more active and self-regulated process developed by students. In this dynamic, participation was not only more frequent, but also necessary, as the lessons required students’ contribution for the development of the class.

Now, while this may appear beneficial, it was also found that certain unintended patterns of participation appeared along the way. At first, in some cases, students’ participation changed its focus on learning to a focus on performance, as students started to prioritize completion of tasks quickly over the quality of their outcomes. This is a possible phenomenon when gamification is approached in the class environment. As Reeve (2012) points out, when the extrinsic goals define students’ engagement in class, the learning process of those who in

fact accomplish these goals can be threatened. This could be seen in the next excerpt from field note 6:

*“Since it is still a competence against others’ time, students may rush and ignore the quality of their task just to hand it in faster than others” (Field note 6, lines 79–81).*

The previous description is in line with the following example where words were misspelled or the sentences constructed had lack of meaning.



I think life is bad because have kids like not have  
food did the person die for this

*(Students 10, 29 and 30 artifact, script for the final project)*

Additionally, the style of participation reinforced by rewards, caused that some students -whether due to their lack of interest in learning or a strong desire to obtain more points- started cheating on their classmates, relying on their answers as a strategy to complete the tasks. As the head teacher describes:

*“Sometimes, when students were not sure of the answer, they preferred to wait for the best students, or the groups that were doing better, to answer first, in order to just copy them.”*

*(Teacher interview, own translation).*

This can be tracked back in the theory as Burke (2014) mentions that “one risk of gamification is that winning the game may become the player goal, rather than playing the game. Then, players may look for loopholes to game the system”, (Burke, 2014, p. 117). An example of this can be portrayed in some of the students’ written outcomes that emerged during the activities, where students were asked to write sentences of their own without many restrictions. Still some students copied the information of their peers identically even when they made mistakes.



|                                      |                                      |                                     |
|--------------------------------------|--------------------------------------|-------------------------------------|
| <b>Student 5 and 25's paper</b><br>  | <b>Student 6 and 28's paper</b><br>  | <b>Student 22 and 7's paper</b><br> |
| <b>Student 10 and 24's paper</b><br> | <b>Student 12 and 21's paper</b><br> |                                     |

(Student artifacts, exercises guide 5 above and exercises guide 4 below)

Overall, the patterns regarding participation showed complex dynamics determined by the implementation. While they increased significantly, they also changed the dynamics of the class and students became more involved. At the same time, for some students this situation ended up being an opportunity to have fun with the only aim of obtaining points and prioritizing the competence against others shifting the initial goal of learning through a reflection process.

## Category 2.2. Cognitive depth and attention span.

In terms of attention, it was found that at first instance, students required external regulation to be attentive during the class explanations. This phenomenon was seen before and during the intervention, where several students preferred to talk among themselves, use their phones, or do activities from other subjects while the teacher was explaining. In an attempt to stop this, external tools, such as whistles or penalties, were used as theory establishes that “controlling aspects of an external event pressures students toward a specific outcome or toward a specific way of behaving”, (Reeve, 2012, p. 156). At first, these tools seemed to be successful as distraction got reduced in mass and more and more students paid more attention during class.

*“The teacher in training used the whistle to call the attention of the students to listen to him again. After this, the students remained in silence and the teacher*

*asked them if they were ready, which was answered with a loud yes.” (Field note 7, lines 51–54).*

Nonetheless the use of these resources created dependence on external control, as every time one of these tools had to be used to gain students’ attention since they could not self-regulate their attention span. This aligns with Reeve’s (2012) claim that “controlling external events diminish intrinsic motivation, whereas noncontrolling external events preserve autonomy and maintain intrinsic motivation.” (p. 156). Indeed, at the end of the intervention, due to the overexposure to these stimuli, students developed a sort of tolerance, where even using the tools that at first worked, lost their impact capturing students’ attention, as it was described in the field notes:

*“He tried in different ways to get students’ attention, from speaking up to using the whistle and even to moving both arms. Nonetheless, he could not get all the class attention, since still there were around 8 students who were not paying attention and were talking among themselves. The head teacher had to do a short intervention to get students’ attention, calling for respect to the teacher in training.” (Field notes 10, lines 24-31).*

Another phenomenon found in terms of attention was that in the practice even when students seemed to be attentive to the explanations, this did not mean that they were comprehending properly the input given. In terms of cognitive depth, students achieved a low level of understanding where most of them were able to use the grammar structures that were taught, but only a few got a critical appropriation of them in use, which overall displayed superficial cognitive processing. As pointed out in this excerpt:

*“The teacher in training realized that the students were too focused on memorizing the grammar structure of the tenses they were studying that they forgot to link this information*

*with the practical use. That way they might know how to write sentences in different tenses but not be conscious of the tense itself.” (Field notes 8, lines 14-18).*

This suggests that due to the dynamics of the phygital gamification where students were mostly required to demonstrate their appropriation of the structures, students prioritized learning the form rather than meaning, which limited their ability to internalize the concepts. In line with this, some students also showed that their attention span was restrained to the completion of activities, where learning the topics for completing the task became more relevant than fully understanding them, as the head teacher pointed out:

*“At times it was just competing for competing, without really being aware of what they were learning or paying attention to the details. In the activity, the teacher in training showed them one exercise, and then by the fourth or fifth exercise, he had to repeat it so they could notice where the mistake was.” (Teacher interview, own translation).*

In conclusion, although at first look, students seemed to be more attentive to the class than before when explanations of the topics were offered, the findings revealed in the end, this situation did not translate into deep cognitive processing where students could use the information provided to produce their own meaningful written and oral output in the target language. In other words, students were in silence paying attention but not necessarily learning in a meaningful way due to their task-oriented focus and forced external regulation for their attention span.

### **Category 2.3. Task involvement.**

One of the pervasive aspects observed during the intervention was emotional engagement that students displayed in the classroom, particularly during the activities that were part of the phygital gamification. During these activities, students openly expressed their

emotions depending on the performance they had on them, the rewards they obtained or the penalties that they eventually got as a consequence of their class behavior. For instance, the emotion of worry appeared when:

*“Students waited patiently and with a little of tension the deliverance of the correct answer by the teacher to know if they had answered correctly or not” (Field note 5, lines 86–88).*

Likewise, other emotions such as joy, excitement, and frustration were identified as well, which are illustrated in the following excerpt:

*“The reactions from the students were diverse, while when one challenge was completed in a wrong way they complained or did not say anything, when a challenge was correct, they celebrated, made noise, or smiled before being graded” (Field note 3, lines 184–188).*

These reactions can be directly linked to what Reeve states as the *relatedness* that students eventually developed in the activities. According to Reeve (2012), students display a “need to establish close emotional bonds and secure attachments with others” (p. 154). When they experience this relatedness, they are involved in responsive, caring and reciprocal interactions that exhibit their emotional connection with the environment. Moreover, since the tasks and resources were designed according to students’ age and interests, these appeared to foster an affective connection that made them be involved with the class, as the head teacher pointed out:

*“From the beginning, the design of those activities was very appealing for the students’ age. So, it was quite meaningful for them to have class with the teacher, because they always looked forward to his class with a lot of enthusiasm.” (Teacher interview, own translation).*

Nonetheless, this involvement was also visible in students' behavior beyond the classroom setting. For instance, in the final project, where students had to deliver a recorded video of a news show based on their school context, some seemed to assume the task seriously and with commitment by including things that were not initially asked by the teacher. This can be reflected in the way some students performed, such as in the 24/7 news video where:

*“The video starts with a short intro animation with the name of the program. After that the student#30 appears sitting on a chair and holding a microphone imitating a tv presenter” (Video artifact analysis 24/7 news transcription)*

And also, in the intentional edition of the video trying to resemble as much as possible as a real news show:



*(Video artifact the students' time, students 15, 8, 16, 2).*

*“The video of this group starts by showing an intro animation of the news program which includes a background soundtrack, after this, the first student appears. Something to notice is that their video is placed over a template of a tv program that apparently, they got from Tik Tok. On this template some subheadings such as “breaking news” and “live news” appear, as well as a*

*blue map of the world imitating the templates that real life news programs use.” (Video artifact the students’ time analysis transcription)*

Finally, this involvement was also visible, in the willingness that students had to develop the activities, since most of them seemed to take their time to analyze what they were asked to and did not answer randomly.

This aspect is directly tied to the agentic dimension that Reeve (2012) describes in his theory, where students get a higher level of commitment with the tasks in their learning process and show authentic interest in them. Students engaged “not only react to learning activities but they proact on them—enriching them (...) (e.g., generating options, communicating preferences), and even creating or requesting the learning opportunity in the first place, rather than merely reacting to them as a given” (Reeve, 2012, p.162). As seen in the field diaries, that cognitive effort was driven by their engagement and desire to win but also bounded to the demands and instructions of the activity:

*“Students were deeply thinking about the correct answer making analysis of it since the 30 seconds seemed not to be enough time for the students to answer and they did not choose any option randomly (Field note 5, lines 80–84).*

### **Category 3. Social interaction**

The third major category which emerged along with the research was social interaction, which makes reference to the way students got related among themselves during the development of the activities. In the context of the research, interaction was not only promoted but also required as most of the activities were designed to be developed either in groups or in pairs with the aim of creating a collaborative environment where students could help each other to find solutions and learn in the process.

This perspective goes in line with Reeve's (2012) who under the principles of the basic needs theory proposed originally by Decy and Ryan (1985) identifies relatedness among individuals as one of the three psychological needs that drives intrinsic interest in students.

Naturally, students have the need to establish social relationships in the environment where they interact, these bonds are characterized by kindness, care, receptivity and reciprocity. Taking this into account, the idea of the classroom as the space where students merely coexist transforms into a social environment where relationships are constructed and constantly negotiated. These interactions shape students' behavior and influence their perceptions of the class, which has implications in their learning process. Bearing this in mind, social interaction was analyzed through cooperation, segregation and individual work.

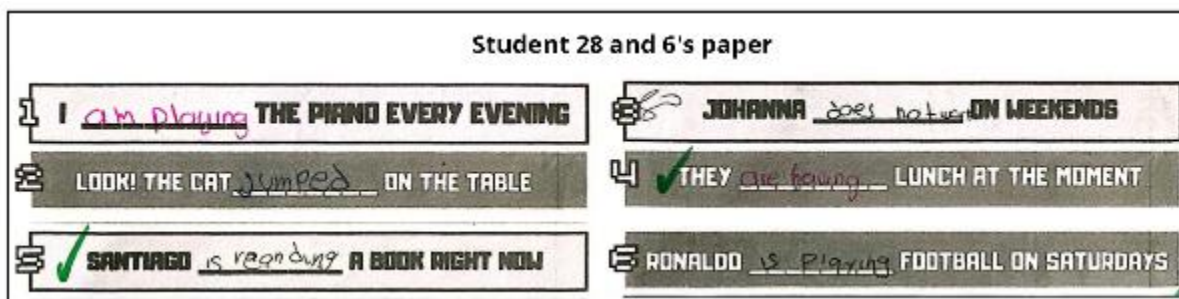
### **Category 3.1. Cooperation**

Firstly, it was observed that over the whole intervention some students participated collaboratively in the completion of tasks, where they shared ideas, supported each other and got agreements to find the answers that they were required to provide. This is a clear manifestation of agentic engagement, where students are expected to be "offering input, personalizing and enriching the lesson, and modifying and adapting it into an improved opportunity for learning" (Reeve, 2012, p. 165). This was observable in the final project where some students got divided in roles and worked collaboratively to create a decent outcome, as described in the video artifact when:

*"The interview section is closed with the conclusions of student#16, the student #15 appears again to present more information. This shows how the project was equally divided and how the students understood the roles each one had to take."*

*(Video artifact from "the students' time" analysis transcription).*

Likewise, cooperation could be seen in some of the written artifacts where there were different types of fonts and ink colors showing that the task was developed by different people and in different moments:



(Student artifacts, exercises guide #4)

This information suggests that the activities proposed created opportunities for group collaboration, allowing students to construct responses with the support of their classmates and learning from the others in the process. However, this pattern was not consistent in all the groups and other ways of interacting appeared consequently. For instance, in some groups students relied on their peers to carry out most of the work, as the field diaries reveal:

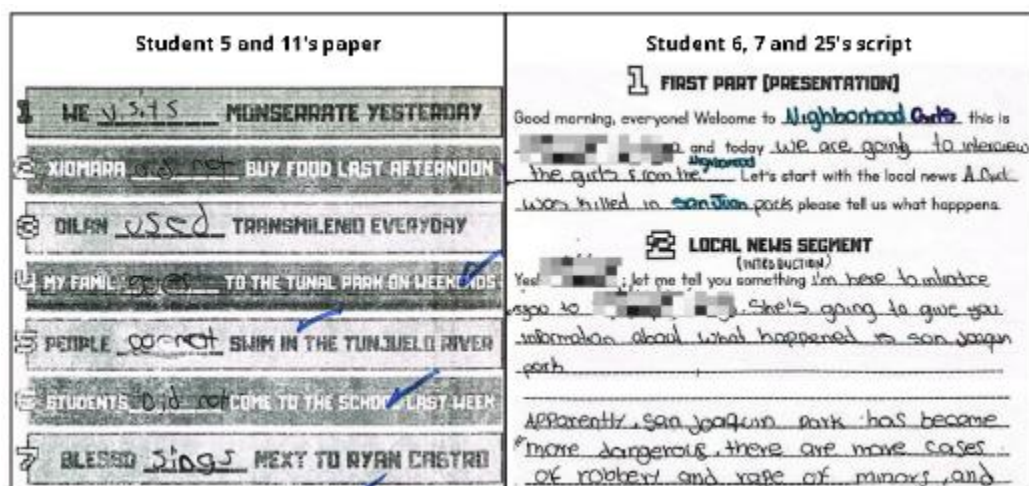
*“During this time the teacher noticed that in some groups some students were not working in a proper manner to be in a group, as only a few of them were writing and completing the task while their peers used their phones to play videogames or to chat on WhatsApp.”*

(Field notes 10, lines 90–93).

### Category 3.2. Individual work

Contrary to what was intended to be completed collaboratively, it was observed that a significant sample of the artifacts showed signs of work carried out independently by only one student of the groups. This was reflected with a series of written answers that followed the exact same font over the whole exercise:





(Student artifacts, exercises guide 3 on the left and final project script on the right)

This aspect was reflected in the creation of the recorded video project, where the members of some groups delegated tasks among themselves, but it was evident that they did not work as a group to deliver the task but preferred to do it individually. In the case of the project 24/ news

*"It was observable the fact that each student decided to do the videos and the parts of it on their own and joined them together without taking care of the coherence it made as a single project" (Video artifact 24/7 news analysis transcription).*

This pattern could be related to the pressure that students created to perform efficiently and their tendency to prioritize completion of activities over discussion and reflection of them. Likewise, this is tied to the design of the phygital gamification, guided by competence and external rewards which were constantly promoted with the aim of increasing motivation in students. In this line, Reeve affirms that "classroom conditions can enhance and support students' intrinsic motivational processes or undermine and thwart them." (2012, p. 156).

Therefore, students may find it more important for the class dynamics to perform correctly to get a high score rather than interacting with their language learning process in an authentic manner, thus, undermining their intrinsic interest in the task. This made them create

strategies to achieve this goal, such as selecting the member of the group with the best performance in English to respond to the exercises on behalf of the rest.

### **Category 3.3. Segregation**

Finally, the findings revealed that not all students participated equally during the activities proposed, which unveiled how segregation may appear when these strategies are taken to unequal environments. As Burke (2014) mentions, gamification “can be highly engaging for the top performers, but may be uninteresting or even demotivating for all but those select few”. (p. 112). In regard to this, the head teacher pointed out that:

*“The good students worked with the good ones, and the average with the average ones, and the weaker ones there off in a corner where they didn’t want to do anything” (Teacher interview, own translation).*

Moreover, students showed the tendency to act according to their own interests and work in groups where there were classmates who had a similar level of English to them, probably thinking that in this way, they would have more possibilities to get a better score. This tendency created a system of hierarchies where the good students kept improving their performances, but the bad ones got stuck in their level, causing more inequalities in the process.

This was visible in several of the exercises, as quickness was a criterion to give a student the opportunity to answer, thus, the students who were not quick raising their hands or giving an answer lost the chance to participate in class, as is described in this excerpt.

*In this moment the reaction of the students intensified drastically and from the 15 couples made for the activity around 11 raised their hands to complete the challenge. This created another problem since the teacher did not know which couple to choose to give them the chance to participate. (Field notes 1, lines 144–148).*

In conclusion, the findings revealed that during the implementation interaction among students was effectively promoted, and in some instances, it led to meaningful ways of cooperation where students shared ideas and worked together in their aim to complete the task while at the same time learned from their peers. Nonetheless, the interaction shown was not equal in some of the groups and what is more, the style of the gamification derived in the creation of differential levels of English performance which fostered inequality in students as well.

#### **Category 4. Linguistic performance**

The fourth and last category identified for this study was linguistic performance, which refers to the students' ability to understand and use the target language in terms of grammar and vocabulary, as Halliday and Matthiessen (2014) affirm, these "are not two separate components of a language" (p. 7). In general terms, although motivation and engagement were fostered, as was outlined in the previous sections, the findings also suggest that this was not necessarily shown in improvements in students' proficiency in the target language. To examine this dimension, three main indicators were considered, comprehension, grammar and vocabulary learning, and oral production.

##### **Category 4.1. Comprehension in English.**

Although the focus of the intervention was on the development of grammar and vocabulary, comprehension in English was considered a relevant subcategory as it was a fundamental component of linguistic performance. Understanding the target language functions as a prerequisite for language use; without it, students are incapable of using grammatical structures and vocabulary items effectively. This goes in line with Halliday's (1979) perspective, who states that "linguistic systems organize in a manner that the social

context provides information to predict the text (...) without this information there would not be communication" (p. 245).

Within this framework, the findings showed that students experienced some difficulties, particularly in listening tasks. When the teacher in training delivered instructions or content in English, students' understanding was low, even at the end of the intervention after the students received the input given in class. This could be seen in some of the field notes. For instance:

*"Since this time the explanations were given in English the students did not understand well what they had to do. This shows that at this point of the intervention students still struggle with the listening comprehension in the foreign language". (Field notes 9, lines 27–30).*

This response was visible in the activities developed during the intervention as well, where some students struggled completing listening tasks, displaying a gap between the exposure they had to the target language and real understanding, as this field note shows:

*"The first four exercises were listening exercises, so the students were in charge of listening carefully to what the head teacher was saying and then answer the exercises (...) in this manner the result was a deficient response from the students, as most of them, around 87% failed answering these exercises." (Field notes 7, lines 55–65).*

As a consequence of this lack of understanding in the target language, instructions and explanations had to be translated into Spanish once they were given in English for students to understand them:

"The teacher in training read first in English and then translated it into Spanish" (Field notes 6, lines 31–33).

This situation led to a dependence on the mother language for the development of the class and at the same time displayed the limitations of the input given in English.

## Category 4.2. Grammar and vocabulary

The data revealed that students' learning of grammar structures and vocabulary in English was certainly unequal. This may be explained by different factors identified in previous sections, such as the emergence of segregation among students' English levels, the appearance of hierarchies, and the displacement of the initial learning goals where some students prioritized their performances to get points over the reflection and significant internalization of the topics seen in class.

For instance, in some groups the application of the vocabulary and grammar taught in class could be observed. These students demonstrated a developed use of grammar and vocabulary which allowed them to construct coherent and meaningful messages in regard to the task they were asked to. expressions like *"I think that they have a lack of discipline because they didn't study"* or *"Next we are going to interview..."* reflect an attempt to use complex structures and to organize ideas logically.

| Student 15, 8,16 and 2's script                                                                                                                                                                                | Student 6, 7 and 26's script                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>(INTERVIEW)</p> <p>Good morning <del>to</del> what do you think about the situation? I think that they have a lack of discipline because didn't study why didn't they do it? Because they are very lazy</p> | <p>(INTERVIEW)</p> <p>Next we are going to interview who was one of the robbery victims. How did you feel about the robbery that happened in that area? How did it happen?</p> |
| <p>(CONCLUSION)</p> <p>In conclusion <sup>that</sup> the students of are not learning anything, since no one is showing interest in the class</p>                                                              | <p>(CONCLUSION)</p> <p>What happened was that was a boy robbing a girl in the jacarandá park, and he put a knife to her neck.</p>                                              |

(Student artifacts, script for the final project)

Additionally, these students displayed a wide lexical range, incorporating words related to the topics mentioned such as *lack of discipline* with *lazy*, and *interest* and *robbery* with *victim*, *park*, and *knife*, respectively. This contributed to a clearer representation of the situation being described.

On the other hand, other students showed limited progress in their internalization of grammar and vocabulary. In these written productions students showed difficulties constructing basic sentences, as their productions lacked correct word order, verb forms, and syntactic coherence. For instance, some expressions such as “*bulling bad for Pipo?*”, “*people not kind be don’t come negative commentary*” or “*because have kids like not have food*” show an absence of control in grammar rules and problems in spelling of words.

Moreover, these artifacts display a limited lexical range, where words are used in an isolated way instead of displaying complete expression. What is more, students have a tendency to directly translate ideas from Spanish into English, which affected coherence in their productions.

| Student 25, 3, 27 and 24's script                                                               | Student 10, 30, and 29's script                                                                   |
|-------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| (INTERVIEW)  | (INTERVIEW)  |
| Bulling bad for Pipo? yes verr very bad bicaez                                                  | I think like is bad because have kids like not have                                               |
| arent peepie                                                                                    | food and the persone die for this                                                                 |
| Have you                                                                                        |                                                                                                   |
| by by you                                                                                       |                                                                                                   |
|                                                                                                 |                                                                                                   |
| (CONCLUSION)                                                                                    | (CONCLUSION)                                                                                      |
| Peepie not kind be, don't come negative commentary, it                                          | the kids <del>are</del> not have food for flow                                                    |
| is verr bad and feels bad for dis                                                               |                                                                                                   |
| Have you so much                                                                                |                                                                                                   |

(Student artifacts, script for the final project)

This perspective is reinforced by the perception of the teacher, who affirmed that

*“There were some students who did not seem solid in their knowledge, different from others who showed a positive progress, but they did not probably because they were obsessed with the style of competence” (Teacher interview, own translation).*

This situation reflected the limitations of learning grammar in isolation, as not all students had the same basis in the target language, some required more input to fully understand

language and then show an outcome within it. As Richards and Renandya suggest (2002), teaching grammar without enough contextual input is insufficient to learn since “when it comes to using language in practice, students discover that they lack vital elements (...) they can recite verbs but not sustain a conversation” (p. 151), highlighting the importance of providing enough contextualized input for effective language use.

#### **Category 4.3. Oral production.**

Finally, in terms of oral production, the findings suggest that students presented difficulties when speaking English, particularly in their pronunciation and fluency. During the class time, students tended to pronounce English when reading as if it was Spanish using the phonemes of this language, which affected the clarity of their speech, as described in the next excerpt:

*the teacher in training monitored from first-hand, plenty of mispronunciations were found in the students, as most of them pronounced the words as if it were Spanish and they doubted whether to pronounce words in one or other way, which led them to have a non-fluent speech (Field notes 11, lines 40–45).*

This was an expected phenomenon, as students did not have many opportunities to refine their speech and precision was not a sought goal from the beginning. Naturally, Thornbury (2005) states that “learners will betray to varying degrees, the influence of their first language pronunciation when speaking a second language, and this need not be a problem so long as intelligibility is not threatened.” (p. 36). On the other hand, some students exhibited a lack of genuine language production, since they preferred to read the whole script in a mechanical way instead of producing speech independently and in a spontaneous way, as it was observed in the final recorded video project of one of the groups:

*“The student seems to be trying to read something behind the camera (...) he gets stuck in what he is saying and gets closer as trying to see clearly what he has to say. (...) He also speaks without confidence as not being aware of what he is saying and he ask a question that he answers to himself which displays their unconsciousness to the task and their lack of preparation” (Video artifact Guillermo News analysis transcription)*

This pattern indicated that students' oral production was in some cases preplanned rather than spontaneous as they depended on the written content that they wrote in advance to communicate their ideas, reflecting a lack of authentic oral communication. Following Thornbury's view (2005) this can be explained by the fact that speaking is an ability that requires not only knowledge of the language but also automatization, as “skill comes by the constant repetition of familiar feats rather than by a few over-bold attempts at feats for which the performer is yet poorly prepared” (p.6). All this together suggested a limited internalization of the language, as students were not able to produce language outcomes independently.

## **Chapter VI: Conclusions**

Along this chapter, the main conclusions of the pedagogical intervention are discussed in relation to the research question, the general, and the specific objectives. Moreover, this includes the pedagogical and context implications for the implementation of this strategy, and the limitations experienced. Finally, suggestions for further research are explored.

### **Conclusions**

In general terms, this study aimed to describe the role that Phygital Gamification played in addressing motivation and engagement towards English learning among 702 students at Guillermo Cano Isaza School. Based on the analysis of the findings it can be affirmed that the intervention mediated with the proposed strategy did have a significant impact fostering motivation and engagement towards learning English within the boundaries



of the classroom setting. However, these two features by themselves did not guarantee meaningful learning of the target language.

To tackle this issue, it is important to refer back to the specific objectives which provided the path for addressing the main objective and answering the research question. In response to the first specific objective, which aimed to identify how Phygital Gamification triggered students' interest in English learning in the class, data showed that through the intervention students developed a more participative disposition and a positive attitude to the class, which made them be more involved in the activities in class.

Some students also gained confidence to learn, since before the intervention they avoided participating in class out of their fear of making mistakes or being judged by others. All together, these findings suggest that in terms of motivation there was a favorable result, considering how low interest was for the class before the intervention due to the socio-cultural value that the language had in the student's context and the methodologies that were carried out by the head teacher.

Nonetheless, some students developed interest in the gamification elements and classroom activities but not in learning the language as a purpose itself, which caused displacement of the intended learning goals. This was a possible consequence of introducing extrinsical rewards to motivate them, which disrupted their intrinsic interest to learn English and transformed it into learning just to gain points, overcome the challenges and obtain a decent grade. All of this implied memorizing only the necessary information to achieve these purposes without reflection of the knowledge acquired.

Considering the second specific objective, where the aim was to examine how Phygital Gamification influenced students' behavioral engagement in the classroom, it can be concluded that through the strategy, a more dynamic, appealing and interactive environment

was created successfully. The students got emotionally involved in the activities proposed, showing regular emotions of joy, excitement, frustration and worry, which reflected the level of immersion and affective connection that they achieved to be involved within the class. Likewise, participation grew as students perceived the value of their comments within the reward system and the construction of the class as well.

On the contrary, some students started cheating on their classmates as a strategy to complete the tasks and benefit from it. This reinforces the idea that although engagement increased significantly, for some students it ended up being an opportunity to have fun with the mere objective of competing and obtaining points, prioritizing the dynamics of the class over the pedagogical purpose of them. What is more, the attention span of students depended on many cases of external control, which at the end was ineffective since the students generated tolerance to the stimuli to capture their attention and did not develop the ability to self-regulate their behavior.

On the other hand, during the intervention, some social interaction variables emerged which were meaningful to consider in the analysis of the data. These findings revealed that interaction among students increased, but it was not always either collaborative or equal. Although some students supported each other in their process and discussions to co-construct knowledge were achieved, it was discovered that major participation occurred individually, where pairs or groups chose the person with the best performance to carry out the activities that were designed to be developed collaboratively. Additionally, students created groups with peers who had a similar level of English to them, broadening the inequality gap in the process.

Now, bearing in mind the third and last specific objective, whose aim was to determine the extent to which Phygital Gamification contributes to students' linguistic

performance in English throughout the intervention period, it can be concluded that the strategy did not achieve the ideal of ensuring language learning as the implementation of it was not sufficient on its own to reach the expected level of English. Although students were constantly exposed to the target language and multiple opportunities to use grammar and vocabulary in context were given, only very specific students presented advancements and progress in their proficiency in English.

As for the rest, they kept showing a low performance and, in many cases, superficial development. In fact, difficulties in listening comprehension and oral production persisted, and genuine use of the language was barely observed in some groups. At the end, while the intervention successfully created conditions for promoting motivation and engagement during the class and transforming the dynamics that students were used to before, it did not sufficiently promote a meaningful internalization of the language. This suggests that when approaching Phygital Gamification complementary pedagogical strategies and time are required to guarantee progress in language learning.

### **Implications**

The first implications that arose from this research were the pedagogical ones. Throughout this project, it was seen that motivation and engagement to the class increased when contextualized and challenging activities were involved. In this way, students developed an emotional connection that allowed them to change their pessimistic perspective of the class and made them more involved in it.

Nonetheless, not all students became conscious of their own learning process, as they mostly focused on their experience in the activities rather than in the learning value in them. Therefore, when approaching gamification of any type as a teaching strategy, it must be considered that its effectiveness does not rely on its ability to engage students, but on how it is pedagogically designed and oriented to direct the strategy towards meaningful learning. In this

line, teachers must provide opportunities for reflection, critical thinking, and explicit use of the target language, hence, students do not lose the purpose of learning the language.

On the other hand, at an institutional level, this research had certain implications, as the findings suggest that the teachers from the school should reconsider some traditional, teacher-centered, and textbook-based practices that are still prevalent and that contribute to the emergence of a hostile attitude from students towards both the class, and the English language. The impact found, indicates that by giving students a more participative role in the class, and adapting activities to their needs, this can contribute to the improvement of the classroom dynamics, as students transform their stance to learn English.

### **Limitations**

In relation to the limitations of this research, these go from aspects related to the dynamics of the school to the design of the strategy itself. As was outlined in the description of the population, the school provided lunch service to the students. The negative issue with this situation was that the schedule for lunch overlapped with class time, and even this time could change depending on when the dishes were prepared in advance or not for students. Therefore, students were required to abandon the classroom at any point of the class to have their lunch, which directly affected the continuity of classes and the concentration span for some students, making it difficult for the teachers to recover the control of the class when students got back to the classroom after they had their lunch.

On the other hand, there were limitations with the methodology. Since the strategy was designed to work in both dimensions, physical and digital, it required the permanent use of elements that, particularly the technological ones, were not present in all the school settings and were subject to present unexpected changes that could alter the normal development of the intervention. For instance, there were some occasions where the

technological devices used such as the tv or the projector did not work due to being damaged or because absence of power in the institution, thus, the teaching processes and class plan had to be changed in order to continue with the lesson.

Finally, another limitation was the language used during class. Even when there was enough input given in the target language, students still required a translation of most of the information given into Spanish. This generated dependence on using the mother tongue and limited the opportunities to learn in a more immersed and natural manner.

### **Further research**

To conclude, this research showed that phygital gamification can increase students' motivation and engagement when approaching it as a teaching strategy within the EFL environment. However, it did not consistently lead to improvements in all students' linguistic performance. As a consequence, future studies could explore how gamified strategies can eventually contribute to deeper language learning and examine relationships between engagement and meaningful learning.

Additionally, it would be relevant to develop this strategy with different age groups and different socioeconomic backgrounds to determine if phygital gamification triggers motivation and engagement independently from the age and the context of the participants and to see if the groups show similar patterns in linguistic performance.

## References

Alizadeh, I. (2016). Vocabulary teaching techniques: A review of common practices. *International Journal of Research in English*, 1(1), 22–30.

Almanza, A. (2024). *Gamification: A tool to improve speaking skill through the TBL approach* [Trabajo de grado, Universidad Pedagógica Nacional]. Repositorio Institucional UPN. <http://hdl.handle.net/20.500.12209/19735>

Aravena, M., Kimelman, E., Micheli, B., Torrealba, L., & Zuñiga, J. (2006). *Investigación educativa I*. Universidad de Los Lagos. <https://hdl.handle.net/20.500.12799/4687>

Braun, V., & Clarke, V. (2021). *Thematic analysis: A practical guide*. SAGE Publications.

Brufee, K. A. (1999). *Collaborative learning: Higher education, interdependence, and the authority of knowledge* (2nd ed.). ERIC.

Burke, B. (2014). *Gamify: How gamification motivates people to do extraordinary things*. Bibliomotion.

Capacity Building Commission. (2023). *Transforming to a phygital world of capacity building*. Government of India.

Casarini, M. (2012). *Teoría y diseño curricular*. Editorial Trillas.

Cerda, H. (1991). *Los elementos de la investigación*. Editorial El Búho Ltda.

Cohen, L., Manion, L., & Morrison, K. (2007). *Research methods in education* (6th ed.). Routledge.

Colegio Técnico CEDID Guillermo Cano Isaza IED. (2016). *Proyecto Educativo Institucional (PEI)*.

Delgado, J., & Chapetón, M. (2015). *Getting the picture of iconography: An innovative way to teach English to young learners*. Aula.

Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). *From game design elements to gamefulness: Defining gamification*. Proceedings of the 15th International Academic MindTrek Conference, 9–15. <https://doi.org/10.1145/2181037.2181040>

Esquivel, L. (2019). *Gamification: A mission to foster students' engagement and interaction in the EFL classroom* [Trabajo de grado, Universidad Pedagógica Nacional]. Repositorio Institucional UPN. <http://hdl.handle.net/20.500.12209/10368>

Evangelou, D., Dobbs-Oates, J., Bagiati, A., Liang, S., & Young Choi, Y. (2009). Talking about artifacts: Preschool children's explorations with sketches, stories, and tangible objects. *Early Childhood Research & Practice*, 12(2).

Fandiño, Y. J. (2014). Teaching culture in Colombia Bilingüe: From theory to practice. *Colombian Applied Linguistics Journal*, 16(1), 81–92.

Goretti, G., Terenzi, B., Cianfanelli, E., Crescenzi, P., Colombo, C., & Civitelli, E. (2020). *A phygital approach to playful experience in learning process for kids with special educational needs*. ResearchGate.

Guerrero, L. (2023). *Gamification, a strategy to enhance speaking skills in a blended learning environment* [Trabajo de grado, Universidad Pedagógica Nacional]. Repositorio Institucional UPN. <http://hdl.handle.net/20.500.12209/18525>

Halliday, M. (1979). *El lenguaje como semiótica social: La interpretación social del lenguaje y del significado*. Fondo de Cultura Económica.

Hine, G. (2013). The importance of action research in teacher education. *Issues in Educational Research*, 23(2), 151–163.

Jociles, M. (2017). La observación participante en el estudio etnográfico de las prácticas sociales. *Revista Colombiana de Antropología*, 54(1), 121–150.

Kalantzis, M., & Cope, B. (2012). *Literacies*. Cambridge University Press.

Khan, M., Khan, I., Din, S., Ismail, H., Rad, K., Khattak, F., & Jan, R. (2015). The impacts of ICT on the students' performance: A review of access to information. *Journal of Education and Practice*, 6(1), 1–7.

Kearsley, G., & Shneiderman, B. (1998). Engagement theory: A framework for technology-based teaching and learning. *Educational Technology*, 38(5), 20–23.

Lepper, M. R., Greene, D., & Nisbett, R. E. (1973). Undermining children's intrinsic interest with extrinsic rewards: A test of the “overjustification” hypothesis. *Journal of Personality and Social Psychology*, 28(1), 129–137. <https://doi.org/10.1037/h0035519>

Maslow, A. H. (1943). A theory of human motivation. *Psychological Review*, 50(4), 370–396. <https://doi.org/10.1037/h0054346>

McNiff, J. (2002). *Action research: Principles and practice*. RoutledgeFalmer.

Mella, O. (1998). *Naturaleza y orientaciones teórico-metodológicas de la investigación cualitativa*. CIDE.



Ministerio de Educación Nacional. (2017). *Estándares básicos de competencia en lenguas extranjeras: Inglés (Guía No. 22)*.

Ojie-Ahamiojie, G. (2024, January 15). *Using collaborative learning to elevate students' educational experiences*. Faculty Focus.

Pinder, C. (1998). *Work motivation in organizational behavior*. Pearson Education.

Ramos, J. L., Cattaneo, A. A. P., de Jong, F. P. C. M., & Espadeiro, R. G. (2022). Pedagogical models for the facilitation of teacher professional development via video-supported collaborative learning: A review of the state of the art. *Journal of Research on Technology in Education*, 54(5), 695–718. <https://doi.org/10.1080/15391523.2021.1911720>

Ramos, L., & Herrada, L. (2023). *Gamification as a tool for practicing speaking skills* [Trabajo de grado, Universidad Distrital Francisco José de Caldas]. Repositorio Institucional RIUD. <http://hdl.handle.net/11349/40402>

Reeve, J. (2012). A self-determination theory perspective on student engagement. In S. L. Christenson, A. L. Reschly, & C. Wylie (Eds.), *Handbook of research on student engagement* (pp. 149–172). Springer.

Richards, J., & Renandya, W. (2002). *Methodology in language teaching: An anthology of current practice*. Cambridge University Press.

Schöbel, S., Janson, A., & Söllner, M. (2020). Capturing the complexity of gamification elements: A holistic approach for analyzing existing and deriving novel gamification designs. *European Journal of Information Systems*, 29(6), 641–668. <https://doi.org/10.1080/0960085X.2020.1796531>

Shpancer, N. (2004). What makes classroom learning a worthwhile experience? *The NEA Higher Education Journal*, 23–35.

Swan, M. (2002). Seven bad reasons for teaching grammar - and two good ones. *English Language Teaching Journal*, 56(2), 148–152.

Thieves, S., & Lins, S. (2014). *Gamifying information systems: A synthesis of gamification mechanics and dynamics*. ResearchGate.

Thornbury, S. (1999). *How to teach grammar*. Longman.

Thornbury, S. (2002). *How to teach vocabulary*. Pearson Education.

Thornbury, S. (2005). *How to teach speaking*. Pearson Education.

Valenzuela, M. Á. (2021). Gamificación para el aprendizaje. *Revista Educación Las Américas*, 11(1), 91–103.

Vargas-Jiménez, I. (2012). La entrevista en la investigación cualitativa: Nuevas tendencias y retos. *Revista Electrónica Calidad en la Educación Superior*, 3(1), 119–139.

Vygotsky, L. S. (1986). *Thought and language* (A. Kozulin, Ed. & Trans.). MIT Press. (Original work published 1934).

## Annexes

### Annex 1 Characterization questionnaire



## Caracterización Escolar Guillermo Cano Isaza

**B I U G X**

El objetivo del presente cuestionario es la recolección de datos personales de la comunidad escolar presente en el grado 702 como parte de un proyecto académico desarrollado en el ámbito institucional de la Universidad Pedagógica Nacional. Es de vital importancia conocer que la información recogida en este cuestionario será para el uso exclusivamente académico y no se compartirá públicamente en ninguna fuente de terceros.

Nombre \*

Texto de respuesta corta

(1) ¿Cuál es tu identidad de género?

- ☐ Masculino
- ☐ Femenino
- ☐ No binario
- ☐ Prefiero no decirlo
- ☐ Otra...

(2) ¿Cuántos años tienes?

Texto de respuesta corta

(9) ¿Te gusta el inglés?

☐ Sí

☐ No

(10) ¿Qué tanto se te facilitan o dificultan estas habilidades en inglés? Valora cada una del 1 al 5, siendo 1 'muy fácil' y 5 'muy difícil'.

|           | 1 (Muy Fácil)            | 2 (Fácil)                | 3 (Es complej...         | 4 (Difícil)              | 5 (Muy Difícil)          |
|-----------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| ESCUCHAR  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| HABLAR    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| LEER      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| ESCRIBIR  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| GRAMÁTICA | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

(11) ¿Consideras que el inglés es importante para tu vida a futuro?

☐ Sí

☐ No

(12) ¿Cómo te sientes durante tu clase de inglés en el colegio? (Puedes marcar más de una opción)

☐ Emocionad@

☐ Interesad@

☐ Confundid@

☐ Frustrad@

☐ Aburrid@

☐ Otra...

## Annex 2 Diagnostic test

|                                                                                   |                         |
|-----------------------------------------------------------------------------------|-------------------------|
| <b>DESAFÍO DE APTITUDES DE INGLÉS GRADO</b>                                       |                         |
|  |                         |
| <b>NOMBRE:</b><br>/NAME/                                                          | <b>FECHA:</b><br>/DATE/ |

1) Read the text about the favorite place of \_\_\_\_\_ after that, answer the questions using the present simple.

/Lee el texto sobre el lugar favorito de Darwin, después, contesta las siguientes preguntas usando el presente simple/

### THE FAVORITE PLACE OF DARWIN

My name is \_\_\_\_\_ and my favorite place to be is the mountains. I like to walk through the roads of the mountains where I can see the tall trees and hear the birds sing. Also, I enjoy to feel the fresh air and to see the beautiful views from the top. Being in the mountains makes me feel at peace and relaxed and that is the reason why I like it so much.

A) What is the favorite place of \_\_\_\_\_ ? \_\_\_\_\_  
/¿Cuál es el lugar favorito de \_\_\_\_\_?/

B) How does he feel when he goes to that place? \_\_\_\_\_  
/¿Cómo se siente él cuando va a ese lugar?/

C) What are the activities that \_\_\_\_\_ does in that place? \_\_\_\_\_  
/¿Qué actividades hace \_\_\_\_\_ en ese lugar?/

D) Do you think he would feel the same if he goes with someone? \_\_\_\_\_  
/¿Crees que él se sentiría igual si va con alguien?/

2) Write a paragraph in English using simple present from at least 4 lines where you describe your favorite place. Try to mention the elements that you like of that place, the activities you do there and the reason why you like it.

/Escribe un párrafo en inglés usando el presente simple de al menos 4 líneas en el que describas tu lugar favorito. Procura nombrar los elementos que más te gustan de ese lugar, las actividades que haces allí y la razón por la que te gusta ese lugar./

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |

3) Listen to the audio and write three sentences about what you remember of the favorite place of Cristiano Ronaldo. You can answer the questions: What is his favorite place?, What does he do there?, Why does he like it?

/Escucha el audio y escribe al menos tres frases de lo que recuerdes sobre el lugar favorito de Cristiano Ronaldo. Puedes responder a las preguntas: ¿Cuál es su lugar favorito?, ¿Qué hace él allí? ¿Por qué le gusta ese lugar?/

|  |  |
|--|--|
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |
|  |  |

### Annex 3 Sample excerpt from field diaries

|                                                                                 |                                                                                                                                        |                       |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Field diary #3                                                                  | Setting: Colegio Tecnico Guillermo Cano Isaza                                                                                          | Date: March 25th 2025 |
| Observer                                                                        | Class: English                                                                                                                         |                       |
| Topic: Daily routines, habits and past memories. Present simple and Past simple | Objective:<br>Student will the structure of present simple and past simple by talking about routines, habits and significant memories. |                       |
| Categories of analysis                                                          |                                                                                                                                        |                       |
| Intrinsic interest                                                              |                                                                                                                                        |                       |
| Readiness to the class                                                          |                                                                                                                                        |                       |
| Exogenous variables                                                             |                                                                                                                                        |                       |
| Participation patterns                                                          |                                                                                                                                        |                       |
| Cognitive Depth and Attention Span                                              |                                                                                                                                        |                       |
| Task involvement                                                                |                                                                                                                                        |                       |
| Comprehension in English                                                        |                                                                                                                                        |                       |
| Grammar and vocabulary                                                          |                                                                                                                                        |                       |
| Oral production                                                                 |                                                                                                                                        |                       |
| Cooperation                                                                     |                                                                                                                                        |                       |
| Segregation                                                                     |                                                                                                                                        |                       |
| Individual work                                                                 |                                                                                                                                        |                       |

|    | OBSERVATION                                                                                                                                                                                                                                                                                                            | INTERPRETATION                                                                                                                                                                           |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | The teacher in training arrived to the classroom at 12:27. Most of the students were already inside the classroom and the head teacher had not arrived yet. The teacher in training started to write on the whiteboard the titles Present simple and Past Simple, while the students talked among themselves.          | Not having electricity could end up in a problem for the development of the classes, since it is fundamental for the use of the Tv and the laptop to implement the tools that create the |
| 2  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 3  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 4  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 5  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 6  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 7  | At 12:35 the head teacher arrived to the classroom, and greeted her students and the teacher in training. Then they two got together next to the desk to turn on the tv and the laptop to begin the class. Unexpectedly, the Tv was not working, because the power outlets, where the wires were plugged, did not have |                                                                                                                                                                                          |
| 8  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 9  |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 10 |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |
| 11 |                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                          |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 12 | energy. So, the teacher in training, told the head teacher that the first part of the class could be developed without the use of technological devices, while the problem was resolved, to develop the following parts which required them.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | gained experience. Probably a plan b is necessary to undertake these situations in a proper way.                                                                                                                                                                                                                                                                                                                                                                                       |
| 13 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 14 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 15 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 16 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 17 | At 12:44 the teacher in training spoke to students, and told them that the class was going to start from that moment, so they have to sit down, remain quiet, and pay attention. From 25 students who attended today's class, around 13 followed the instruction, while other 12 kept doing what they were doing as they had not listened nor paid attention. In this moment the head teacher speaks loudly to the students and asks them to be quiet and listen carefully to the teacher, she also asks her students why they are not paying attention to the teacher. Then, no one answers.                                                                                                                                                                                                                                                                                                                                                                              | Probably the students do not see the teacher in training as a figure of authority and that is the reason why they do not obey instructions in the intended way. The reasons why they would not see as someone to obey could be that the teacher in training's appearance is not as from an adult, or that does not speak as loud as needed. Other interpretation could be that the teacher in training has not spent as much time with them as for them to consider him "the teacher". |
| 18 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 19 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 20 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 21 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 22 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 23 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 24 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 25 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 26 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 27 | Having the attention of the students, the teacher in training started to ask about what the students remembered of the topic present simple and past simple. Some students started to participate, and the teacher stopped them saying that they could only participate by raising their hand, and waiting for the teacher to give them the chance to talk. Then three students raised their hand to participate and the teacher gave them voice. Among their answers, they mentioned that present simple was used for things that occur constantly and for routines, while past simple was used for events that already happened. The teacher wrote this information on the whiteboard and then he gave diamonds to the people who participated. After this, the teacher asked his students to remember and say which were the component elements of a sentence. Over 5 students raised their hands and the teacher gave voice to one of them to mention it. This student | Apparently, the students had not had the opportunity to have a class created from their participation, or if they had it, they did not conceive the notion to participate by turns. This could be a starting point for creating a notion of "control" into the class, by mandatory rules. A space where the students can be active and take part into the class under the instructions that the teacher gives.                                                                         |
| 28 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 29 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 30 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 31 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 32 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 33 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 34 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 35 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 36 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 37 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 38 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 39 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 40 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 41 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| 42 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |



#### Annex 4 Questionnaires about student's perception

**NOMBRE Y APELLIDO:**

1) ¿Cómo te sientes aprendiendo inglés con Diamantes y Word Warriors?  
(Marca una sola opción)

☐ Aburrido ☐ Confundido ☐ Normal ☐ Emocionado ☐ Motivado



**NOMBRE Y APELLIDO:**

1) ¿Consideras que las estrategias de gamificación, te motivan a prestar más atención en clase y aprender?  
(Responde Sí o No y Por qué)

---



---



---

2) ¿Consideras que usar diamantes en la clase, te motiva a participar y estar más activo de esta? (Responde Sí o No y Por qué)

---



---



---

3) ¿Cómo enmarcarías el uso del silbato para manejar los tiempos de la clase y retomar la atención de los estudiantes? (Tacha con X una sola opción)

☐ Molesto ☐ Normal, no me afecta ☐ Útil, me ayuda a retomar la atención ☐ Chistoso



NOMBRE Y APELLIDO: /

STUDENT 8

1) Describe con tus propias palabras qué fue lo que más te gustó de las actividades del profe (por ejemplo: los diamantes, el avatar, los desafíos de responder la guía y moverse a las cajas, el proyecto del noticiero).

Me gusta ser un actor.

2) ¿Qué fue lo que menos te gustó o que te pareció raro?

nada

3) ¿Cuando recibías diamantes (puntos) sentías que era igual a sacar una nota habitual?

No, es más emocionante sentir la tensión

4) ¿Sentiste que aprendiste más inglés con estas dinámicas que con clases "normales"? Explica con un ejemplo

Si, la verdad en las clases me ayudaron a entender el presente continuo

5) ¿Crees que estas actividades mejoraron o empeoraron tu concentración y comportamiento en clase? Da un ejemplo

Lo mejoraron me volvi más atenta

6) ¿Cómo te llevaste con tus compañeros en estas actividades (competencia, ayuda, conversaciones, peleas)?

Buen me ayudaron mucho

7) ¿Te gustaría que estas actividades se sigan usando en la clase de inglés? ¿Por qué sí o por qué no?

Si, por que hacen el inglés más fácil de aprender, comprender y conjugarse



## *Annex 5 Sample of the transcription of the semi structured interview with the head teacher*

**Entrevistador:** Sí. Entonces, eh vamos a comenzar con la primera pregunta. Y es la siguiente. Eh, desde su experiencia como docente acompañante, ¿cómo describiría el desarrollo general de las clases durante la implementación del proceso de formación?

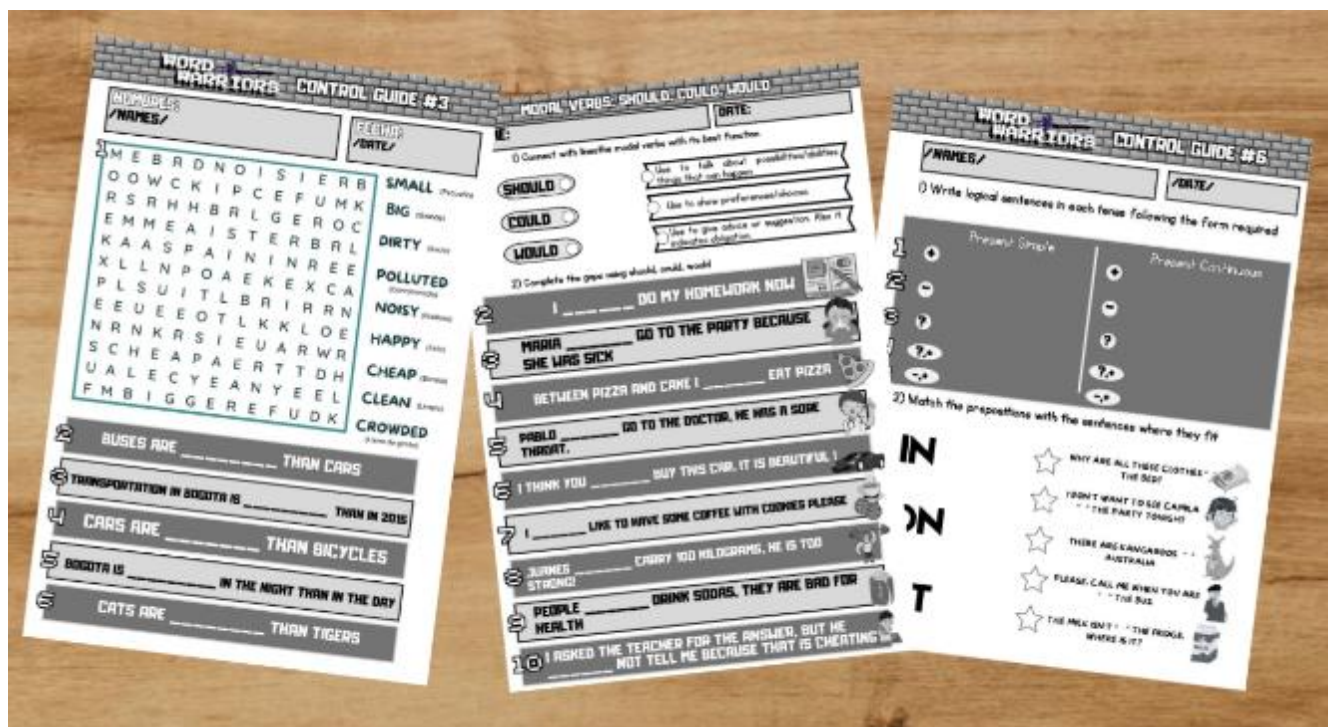
**Entrevistado:** Pues eh Bueno, fue un eh profesor practicante eh muy organizado. Siempre tenía muy bien claro cuál era la malla curricular, cuáles eran los contenidos eh que debíamos abordar en cada uno de los periodos académicos con los estudiantes. Entonces, siempre fue muy cuidadoso con la metodología y las actividades que diseñaba para que además de apuntarle a la práctica misma del contenido o de los temas que se necesitaban que fuera como muy gradual. Entonces, había siempre como un esquema, un orden de actividades. No era así por hacerlo, sino eh tenía siempre una metodología, un orden para desarrollarlas. Y algo que impactó muchísimo fue la gamificación. Desde el comienzo fue muy atractivo el diseño de esas actividades de acuerdo a la edad de los estudiantes. Entonces, fue bastante eh significativo para los estudiantes el tener la clase con el profe porque eh siempre esperaban la clase con mucho gusto. Había estudiantes que conmigo no hm no conectaban con la clase, mientras que con si súper. Entonces, como que estaban siempre ansiosos por esa clase y cuando no cuando él no podía, que fueron muy pocas veces ,creo que fue una vez, tal vez. Eh, como que se sentía como el vacío, como que hacía falta esa intervención.

**Entrevistador:** Listo, profe. Vamos con la siguiente. Y es, si notó algún cambio en la relación interés o actitud de los estudiantes hacia el inglés antes y después de la intervención.

**Entrevistado:** Bueno, antes eh siempre en estos grupos tenemos eh eso es era un grado octavo y pues eh están en están en la época de la adolescencia, entonces eh juegan muchos factores personales, convivenciales, eh están creciendo un poco más, entonces también ellos necesitan saber qué les gusta, qué no les gusta, eh hay resistencia frente al inglés, algunos sí se dejan llevar y les fascina. Hay otros que, bueno, digamos lo que sí, que hay que hacerlo porque es una materia más y hay otros que definitivamente no quieren hacer nada sino decir, "No quiero, no puedo." y no no no se no se comprometen. Entonces eh eso era antes de la intervención. Una vez eh llega ... a la a la intervención eh con con sus actividades así muy llamativas, entonces eh sienten ellos como como que hay una manera de unificar de pronto sus expectativas frente al desarrollo del inglés. Entonces ya se menos quiénes son los que no quieren. Ya se baja el porcentaje muchísimo. Ya no es solo por que no quieren, sino porque eh les gusta. Entonces le ven algo de de de propósito, de significado, de objetivo, de eh utilidad de el aprender el inglés. Entonces eso fue lo que que noté con el antes y el después de la intervención.

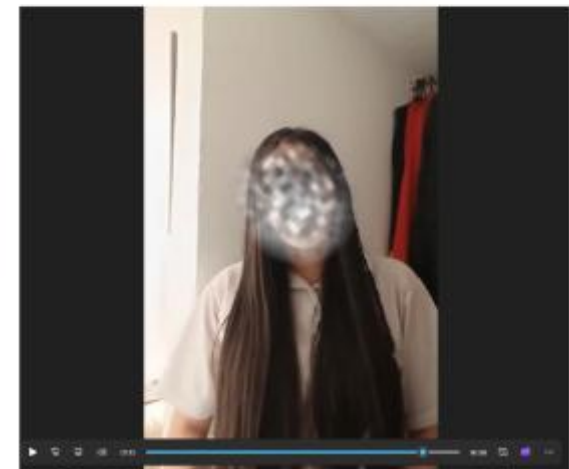
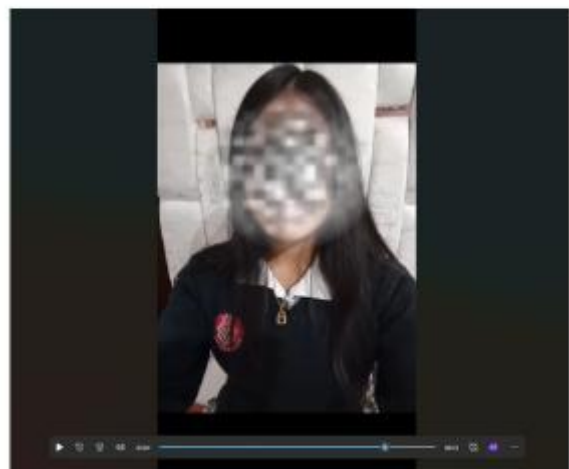
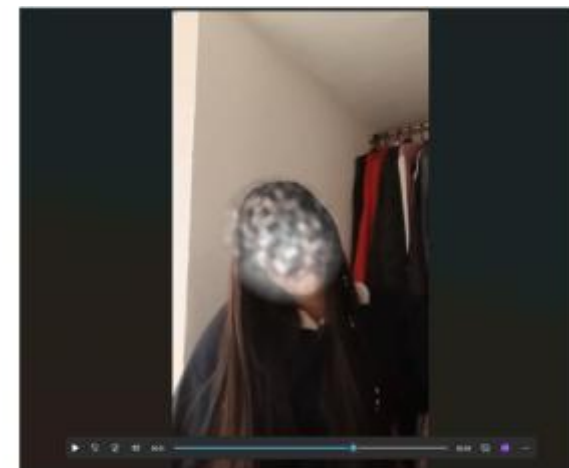
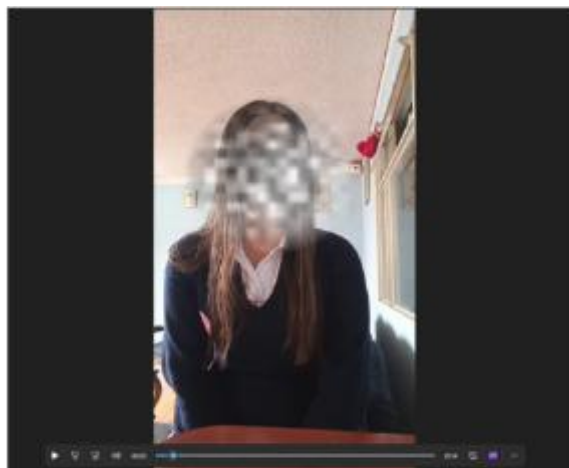


## Annex 6 Student artifacts









## Annex 7 Informed consent

|                                                                                                                                                                     |                                                                                                                               |  |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--|
| <br>UNIVERSIDAD PEDAGÓGICA NACIONAL<br><small>APROBACIÓN DE CONSENTIMIENTO</small> | <b>FORMATO</b>                                                                                                                |  |
|                                                                                                                                                                     | <b>CONSENTIMIENTO INFORMADO PARA LA PARTICIPACIÓN EN INVESTIGACIONES</b><br><b>ADULTO RESPONSABLE DE NIÑOS Y ADOLESCENTES</b> |  |
| Código: FOR025INV                                                                                                                                                   | Versión: 01                                                                                                                   |  |
| Fecha de Aprobación: 02-06-2016                                                                                                                                     | Página 1 de 3                                                                                                                 |  |

**Vicerrectoría de Gestión Universitaria**  
**Subdirección de Gestión de Proyectos – Centro de Investigaciones CIUP**  
**Comité de Ética en la Investigación**

En el marco de la Constitución Política Nacional de Colombia, la Ley 1098 de 2006 – Código de la Infancia y la Adolescencia, la Resolución 0546 de 2015 de la Universidad Pedagógica Nacional y demás normatividad aplicable vigente, considerando las características de la investigación, se requiere que usted lea detenidamente y si está de acuerdo con su contenido, exprese su consentimiento firmando el siguiente documento:

### **PARTE UNO: INFORMACIÓN GENERAL DEL PROYECTO**

|                                                                                  |                                  |                  |  |
|----------------------------------------------------------------------------------|----------------------------------|------------------|--|
| <b>Facultad, Departamento o Unidad Académica</b>                                 |                                  |                  |  |
| <b>Título del proyecto de investigación</b>                                      |                                  |                  |  |
| <b>Descripción breve y clara de la investigación</b>                             |                                  |                  |  |
| <b>Descripción de los posibles riesgos de participar en la investigación</b>     |                                  |                  |  |
| <b>Descripción de los posibles beneficios de participar en la investigación.</b> |                                  |                  |  |
| <b>Datos generales del investigador principal</b>                                | <b>Nombre(s) y Apellido(s) :</b> |                  |  |
|                                                                                  | <b>N° de Identificación:</b>     | <b>Teléfono:</b> |  |
|                                                                                  | <b>Correo electrónico:</b>       |                  |  |
|                                                                                  | <b>Dirección:</b>                |                  |  |

### **PARTE DOS: CONSENTIMIENTO INFORMADO**

Yo \_\_\_\_\_ mayor de edad, identificado con Cédula de Ciudadanía N° \_\_\_\_\_ de \_\_\_\_\_, con domicilio en la ciudad de \_\_\_\_\_ Dirección: \_\_\_\_\_ Teléfono y N° de celular: \_\_\_\_\_ Correo electrónico: \_\_\_\_\_

|                                                                                                                      |                                                                                                                               |  |
|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|--|
| <br>UNIVERSIDAD PEDAGÓGICA NACIONAL | <b>FORMATO</b>                                                                                                                |  |
|                                                                                                                      | <b>CONSENTIMIENTO INFORMADO PARA LA PARTICIPACIÓN EN INVESTIGACIONES</b><br><b>ADULTO RESPONSABLE DE NIÑOS Y ADOLESCENTES</b> |  |
| Código: FOR025INV                                                                                                    | Versión: 01                                                                                                                   |  |
| Fecha de Aprobación: 02-06-2016                                                                                      | Página 2 de 3                                                                                                                 |  |

Como adulto responsable del niño(s) y/o adolescente (s) con:

| Nombre(s) y Apellidos: | Tipo de Identificación | N°    |
|------------------------|------------------------|-------|
| _____                  | _____                  | _____ |
| _____                  | _____                  | _____ |

Autorizo expresamente su participación en este proyecto y

**Declaro que:**

1. He sido invitado(a) a participar en el estudio o investigación de manera voluntaria.
2. He leído y entendido este formato de consentimiento informado o el mismo se me ha leído y explicado.
3. Todas mis preguntas han sido contestadas claramente y he tenido el tiempo suficiente para pensar acerca de mi decisión de participar.
4. He sido informado y conozco de forma detallada los posibles riesgos y beneficios derivados de mi participación en el proyecto.
5. No tengo ninguna duda sobre mi participación, por lo que estoy de acuerdo en hacer parte de esta investigación.
6. Puedo dejar de participar en cualquier momento sin que esto tenga consecuencias.
7. Conozco el mecanismo mediante el cual los investigadores garantizan la custodia y confidencialidad de mis datos, los cuales no serán publicados ni revelados a menos que autorice por escrito lo contrario.
8. Autorizo expresamente a los investigadores para que utilicen la información y las grabaciones de audio, video o imágenes que se generen en el marco del proyecto.
9. Sobre esta investigación me asisten los derechos de acceso, rectificación y oposición que podré ejercer mediante solicitud ante el investigador responsable, en la dirección de contacto que figura en este documento.

Como adulto responsable del menor o adolescente autorizo expresamente a la Universidad Pedagógica Nacional utilizar sus datos y las grabaciones de audio, video o imágenes que se generen, que reconozco haber conocido previamente a su publicación en: \_\_\_\_\_

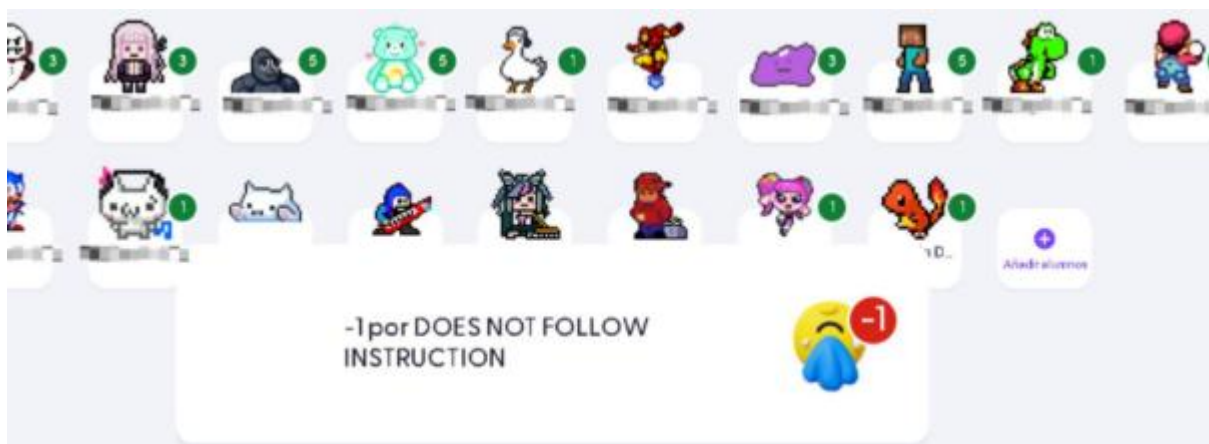
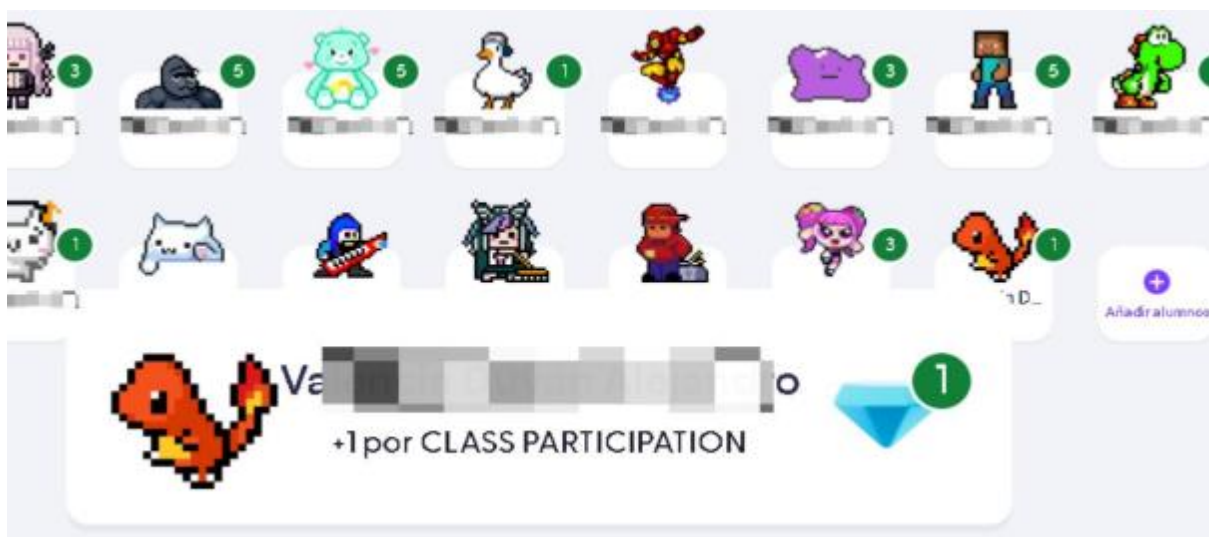
En constancia, el presente documento ha sido leído y entendido por mí, en su integridad de manera libre y espontánea. Firma el adulto responsable del niño o adolescente,

\_\_\_\_\_  
 Nombre del adulto responsable del niño o adolescente: \_\_\_\_\_  
 N° Identificación: \_\_\_\_\_ Fecha: \_\_\_\_\_

Firma del Testigo:

\_\_\_\_\_  
 Nombre del testigo: \_\_\_\_\_  
 N° de identificación: \_\_\_\_\_  
 Teléfono: \_\_\_\_\_

## Annex 8 Digital tools







## *Annex 9 Physical tools*

